Operation and Service Manual

Dual Lock-In Amplifier

SR835

Stanford Research Systems

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Certification

Stanford Research Systems certifies that this product met its published specifications at the time of shipment.

Warranty

This Stanford Research Systems product is warranted against defects in materials and workmanship for a period of one (1) year from the date of shipment.

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General Information

Safety and Preparation for Use

WARNING

Dangerous voltages, capable of causing injury or death, are present in this instrument. These voltages can persist for many minutes after AC power is removed. Do not remove the product covers or panels. Do not apply power or operate the product without all covers and panels in place.

AC Line Voltage

The SR835 Dual Lock-In Amplifier operates from a 90 to 132 VAC or 175 to 264 VAC power source having a line frequency between 47 and 63 Hz.

Power Entry Module

A power entry module, labeled 'AC POWER' on the back panel of the SR835, provides connection to the power source and to a protective ground. The line fuse is internal to the instrument and may not be serviced by the user.

Power Cord

The SR835 Dual Lock-In Amplifier has a detachable, three-wire power cord for connection to the power source and to a protective ground. The chassis of the instrument is connected to the outlet ground to protect against electrical shock. Always use an outlet which has a properly connected protective ground.

BNC shields are connected to the chassis ground and the AC power source ground via the power cord. Do not apply any voltage to the shield.

Service

The SR835 Dual Lock-In Amplifier does not have any user serviceable parts inside. Refer service to a qualified technician. Do not install substitute parts or perform any unauthorized modifications to this instrument. Contact the factory for instructions on how to return the instrument for authorized service and adjustment.

SR835 Specifications

Signal Input

Specification	Value
Input configuration	Single-ended (A), differential (A-B), or current (I)
Input impedance, voltage (DC coupled)	10 MOhm || 25 pF
Input impedance, voltage (AC coupled)	10 MOhm || 25 pF; high-pass f0 = 0.16 Hz
Current input impedance to virtual ground	1 kOhm
Current transimpedance gain options	100 kOhm, 1 MOhm, 10 MOhm, 100 MOhm
Current input bandwidth (100k / 1M / 10M / 100M)	3 MHz / 70 kHz / 7 kHz / 700 Hz
Input voltage noise	2.5 nV/sqrt(Hz) at 1 kHz (gain x1, typical)
Input current noise (1 MOhm / 100 MOhm)	130 fA/sqrt(Hz) / 13 fA/sqrt(Hz) at 1 kHz
CMRR (DC coupled)	>100 dB at DC–10 kHz; −6 dB/octave above 10 kHz
Gain accuracy	+/-1% below 100 kHz; +/-2% to 1 MHz
Gain stability	<50 ppm/deg C
Harmonic distortion	<−80 dBc below 100 kHz; <−70 dBc above 100 kHz
Preamp gain steps	1, 2, 4, 8, 16, 32, 64, 128, 256 (binary, relay-switched)
Line notch attenuation	>=30 dB at 50/60 Hz and 100/120 Hz
Bandpass filter Q	5, auto-tracking

Reference Channel

Specification	Value
Internal frequency range	1 mHz to 1 MHz
Internal frequency accuracy	25 ppm + 30 uHz
Internal frequency resolution	6 digits or 0.1 mHz (whichever is greater)
External reference frequency range	0.5 Hz to 1 MHz (sine or TTL)
External reference, minimum sine amplitude	400 mV peak-to-peak
External reference input impedance	1 MOhm (fixed)

PLL acquisition time	max(2 cycles + 5 ms, 40 ms)
Phase offset range	+/-360 000 deg (displayed wrapped to +/-180 deg)
Phase resolution	360/2^32 deg = 8.4 x 10^−8 deg
Phase noise (internal reference)	<0.0001 deg rms at 1 kHz (100 ms TC, 12 dB/oct)
Phase noise (external TTL reference)	<0.001 deg rms at 1 kHz (100 ms TC, 12 dB/oct)
Phase drift below 20 kHz	<0.002 deg/deg C
Phase drift below 200 kHz	<0.02 deg/deg C
Phase drift below 1 MHz	<0.2 deg/deg C
Orthogonality (X vs. Y)	90 deg +/- 0.001 deg
Absolute phase error	<1 deg above 10 Hz
Harmonic detection range	N = 1 to 99; N x f <= 1 MHz

Demodulator and Filters

Specification	**Value**
Harmonic rejection	−80 dBc
Time constant range	1 us to 86 400 s; any value, 1-2-5 per decade stepping
Filter slopes	6, 12, 18, 24 dB/octave
ENBW—6 dB/oct	1/(4T)
ENBW—12 dB/oct	1/(8T)
ENBW—18 dB/oct	3/(32T)
ENBW—24 dB/oct	5/(64T)
Settling to 99% (6/12/18/24 dB/oct)	5T / 7T / 9T / 10T
Synchronous filter rejection	>=80 dB at N x f_det; always active, full range
DC output stability (digital)	No drift
DC output stability (Aux DAC)	<5 ppm/deg C

Sine Output

Specification	**Value**
Frequency range	1 mHz to 1 MHz
Amplitude range	1 nVrms to 2 Vrms (differential, open circuit)
Amplitude into 50 Ohm	Half open-circuit value
Amplitude accuracy	+/-1%
Amplitude stability	<50 ppm/deg C

Harmonic distortion	<−80 dBc below 100 kHz; <−70 dBc above 100 kHz
Output impedance	50 Ohm (differential)
DC offset range	+/-5 V
DC offset resolution	0.1 mV
Maximum output (|DC| + V_peak)	+/-6 V
Output anti-alias filter	7th-order Cauer LPF, Fc = 9 MHz; −100 dB by 19 MHz

Auxiliary Outputs

Specification	**Value**
Number of channels	2 (Aux-1, Aux-2)
Voltage range	+/-10.500 V
Resolution	1 mV (12-bit DAC)
Output impedance	<50 Ohm
Output current limit	10 mA

Remote Interfaces

Specification	**Value**
USB	USB 2.0 device, USBTMC class
Ethernet	100BASE-T; VXI-11 + TCP socket (default port 5025)
Command format	SCPI (IEEE 488.2 syntax)
Input buffer	256 characters
Output buffer	256 characters
Non-volatile setup slots	9
Maximum streaming rate	1.25 MHz (Ethernet)
Streaming data formats	Float32 (4 B/sample), Int16 (2 B/sample)
Trigger-to-first-sample latency	<=2 ms

10 MHz Timebase

Specification	Value
Timebase In	1 Vrms, 10 MHz; input impedance 50 Ohm
Timebase Out	1 Vrms, 10 MHz; output impedance 50 Ohm
Phase-lock range	+/-50 ppm pull-in range
Multi-unit synchronization	<1 ps rms (typical) of shared 10 MHz

General

Specification	Value
Mains input voltage	100 / 120 / 220 / 240 VAC +/-10%, switch-selectable
Mains frequency	50 or 60 Hz
Power consumption	60 W typical
Fuse (100/120 V)	1 A, 3AG slow-blow
Fuse (220/240 V)	0.5 A, 3AG slow-blow
Operating temperature	+10 to +40 deg C; specs apply +18 to +28 deg C
Storage temperature	−25 to +65 deg C
Operating humidity	<90% non-condensing
Storage humidity	<95% non-condensing
Form factor	19″ rack, 2U (bench use with handles)
Width	432 mm (17″)
Height	133 mm (5.25″)
Depth	432 mm (17″)
Weight	12 kg (26 lbs)

SR835 Command Reference

Common Commands (IEEE 488.2)

Command	Description
`*IDN?`	Returns manufacturer, model, serial, firmware version
`*RST`	Reset all parameters to factory defaults
`*OPC?`	Returns `1` when all pending operations complete
`*CLS`	Clear status registers and error queue
`*TST?`	Execute self-test; returns `0` (pass) or error code
`*SAV <n>`	Save current state to slot n (1–9)
`*RCL <n>`	Recall state from slot n (1–9)
`*ESE <mask>`	Set Standard Event Status Enable register (0–255)
`*ESE?`	Query Standard Event Status Enable register
`*ESR?`	Query and clear Standard Event Status Register
`*SRE <mask>`	Set Service Request Enable register (0–255)
`*STB?`	Query Status Byte register
`*PSC <b>`	Set Power-on Status Clear flag (`0` or `1`)

INPut Subsystem

Command	Description
`INPut:CHANnel<n>:MODE <m>`	Set input topology (`A`, `AB`, `I`)
`INPut:CHANnel<n>:MODE?`	Query input mode
`INPut:CHANnel<n>:COUPling <c>`	Set input coupling (`AC`, `DC`)
`INPut:CHANnel<n>:COUPling?`	Query coupling
`INPut:CHANnel<n>:SOURce <s>`	Select In-1/In-2 (front) or In-3/In-4 rear (`FRONT`, `REAR`)
`INPut:CHANnel<n>:SOURce?`	Query input source
`INPut:CHANnel<n>:CURRent:GAIN <g>`	Set current transimpedance gain (`100K`, `1MEG`, `10MEG`, `100MEG`)
`INPut:CHANnel<n>:CURRent:GAIN?`	Query current transimpedance gain

`INPut:IN<k>:SHIeld <g>`	Set shield/guard connection for input k (`FLOat`, `GROund`)
`INPut:IN<k>:SHIeld?`	Query shield connection for input k
`INPut:IN<k>:GAIN <g>`	Set hardware preamp gain for input k (1, 2, 4, 8, 16, 32, 64, 128, 256)
`INPut:IN<k>:GAIN?`	Query preamp gain for input k

SENSe Subsystem

Command	Description
`SENSe:CHANnel<n>:TC <t>`	Set time constant (1US to 86400S)
`SENSe:CHANnel<n>:TC?`	Query time constant (seconds)
`SENSe:CHANnel<n>:SLOPe <s>`	Set filter slope in dB/oct (`6`, `12`, `18`, `24`)
`SENSe:CHANnel<n>:SLOPe?`	Query filter slope
`SENSe:CHANnel<n>:FILTer:NOTCH <f>`	Set line notch filter(s) (`OFF`, `LINE`, `2LINE`, `BOTH`)
`SENSe:CHANnel<n>:FILTer:NOTCH?`	Query notch filter setting
`SENSe:CHANnel<n>:FILTer:BPASs <b>`	Enable/disable auto-tracking bandpass filter (`ON`, `OFF`)
`SENSe:CHANnel<n>:FILTer:BPASs?`	Query bandpass filter state
`SENSe:CHANnel<n>:PHASe <p>`	Set phase offset (-360000 to +360000 degrees)
`SENSe:CHANnel<n>:PHASe?`	Query phase offset in degrees
`SENSe:CHANnel<n>:PHASe:UNIT <u>`	Set phase unit (`DEG`, `MDEG`, `UDEG`, `RAD`, `MRAD`, `URAD`)
`SENSe:CHANnel<n>:PHASe:AUTO`	Execute Auto Phase (one-shot)
`SENSe:CHANnel<n>:ADCLevel?`	Query ADC utilization (1–5, five 20% bands)

REFerence Subsystem

Command	Description
`REFerence:CHANnel<n>:SOURce <s>`	Set reference source (`INTernal`, `EXTernal`, `DUAL`)
`REFerence:CHANnel<n>:SOURce?`	Query reference source
`REFerence:CHANnel<n>:TRIGger <t>`	Set external reference trigger mode (`SINE`, `POS`, `NEG`)
`REFerence:CHANnel<n>:TRIGger?`	Query trigger type
`REFerence:CHANnel<n>:SHIeld <g>`	Set Ref In BNC shield connection (`FLOat`, `GROund`)
`REFerence:CHANnel<n>:SHIeld?`	Query Ref In shield connection

`REFerence:CHANnel<n>:HARMonic <h>`	Set detection harmonic (1–99, N×f <= 1 MHz)
`REFerence:CHANnel<n>:HARMonic?`	Query harmonic
`REFerence:CHANnel<n>:DUAL:HARMonic <h>`	Set harmonic for f_ext in Dual mode (1–99, N×f_ext <= 1 MHz)
`REFerence:CHANnel<n>:DUAL:HARMonic?`	Query dual harmonic
`REFerence:CHANnel<n>:FREQuency?`	Query actual locked reference frequency (read-only)
`REFerence:CHANnel<n>:UNLock?`	Returns `1` if PLL unlocked, `0` if locked

SOURce Subsystem (Sine Output)

Command	Description
`SOURce:CHANnel<n>:FREQuency <f>`	Set internal oscillator frequency (1 mHz–1 MHz)
`SOURce:CHANnel<n>:FREQuency?`	Query frequency (Hz)
`SOURce:CHANnel<n>:VOLTage <v>`	Set sine output amplitude, differential open circuit (1 nV–2 Vrms)
`SOURce:CHANnel<n>:VOLTage?`	Query amplitude (Vrms)
`SOURce:CHANnel<n>:OFFSet <v>`	Set sine DC offset (-5 V to +5 V)
`SOURce:CHANnel<n>:OFFSet?`	Query DC offset (V)
`SOURce:CHANnel<n>:SHIeld <g>`	Set sine output BNC shield connection (`FLOat`, `GROund`)
`SOURce:CHANnel<n>:SHIeld?`	Query sine output shield connection
`SOURce:CHANnel<n>:PRESet:FREQuency <slot>,<f>`	Store frequency preset (slot 1–6; f 1 mHz–1 MHz)
`SOURce:CHANnel<n>:PRESet:FREQuency? <slot>`	Recall frequency preset (slot 1–6)
`SOURce:CHANnel<n>:PRESet:VOLTage <slot>,<v>`	Store amplitude preset (slot 1–6; v 1 nV–2 Vrms)
`SOURce:CHANnel<n>:PRESet:VOLTage? <slot>`	Recall amplitude preset (slot 1–6)
`SOURce:CHANnel<n>:PRESet:OFFSet <slot>,<v>`	Store DC offset preset (slot 1–6; v −5 V–+5 V)
`SOURce:CHANnel<n>:PRESet:OFFSet? <slot>`	Recall DC offset preset (slot 1–6)
`SOURce:CHANnel<n>:PRESet:LOAD <slot>`	Atomically apply frequency + amplitude + DC preset (1–6)

OUTPut Subsystem

Command	Description

`OUTPut:CHANnel<n>:QUANtity <q>`	Set display quantity (`X`, `Y`, `R`, `THETa`, `XNOise`, `YNOise`, `FREQuency`)
`OUTPut:CHANnel<n>:QUANtity?`	Query display quantity
`OUTPut:CHANnel<n>:REL:ENAble <b>`	Enable/disable REL mode (`ON`, `OFF`)
`OUTPut:CHANnel<n>:REL:ENAble?`	Query REL mode
`OUTPut:CHANnel<n>:REL:CLEar`	Clear stored REL baseline
`READ:CHANnel<n>?`	Return X, Y, R, theta as comma-separated values (synchronous)
`READ:CHANnel<n>:X?`	Return X only
`READ:CHANnel<n>:Y?`	Return Y only
`READ:CHANnel<n>:R?`	Return R
`READ:CHANnel<n>:THETa?`	Return theta (degrees)
`READ:CHANnel<n>:XNOise?`	Return X noise spectral density (V/sqrt(Hz))
`READ:CHANnel<n>:YNOise?`	Return Y noise spectral density
`READ:CHANnel<n>:FREQuency?`	Return locked reference frequency (Hz)
`SNAP?`	Return X_A, Y_A, R_A, theta_A, X_B, Y_B, R_B, theta_B simultaneously

AUXiliary Subsystem

Command	Description
`AUXiliary:OUTPut<m>:VOLTage <v>`	Set Aux output voltage in Fixed mode (m=1,2; −10.5 V to +10.5 V)
`AUXiliary:OUTPut<m>:VOLTage?`	Query programmed voltage (m=1 or 2)
`AUXiliary:OUTPut<m>:MODE <mo>`	Set Aux output mode (`FIXed`, `MEASurement`, `SCAN`)
`AUXiliary:OUTPut<m>:MODE?`	Query mode (m=1 or 2)
`AUXiliary:OUTPut<m>:MEASure:SOURce <s>`	Set measurement source (`XA`, `YA`, `RA`, `TA`, `XB`, `YB`, `RB`, `TB`)
`AUXiliary:OUTPut<m>:MEASure:SOURce?`	Query measurement source
`AUXiliary:OUTPut<m>:MEASure:FSCale <v>`	Set full-scale value mapping to ±10 V output (positive numeric)
`AUXiliary:OUTPut<m>:MEASure:FSCale?`	Query full-scale value
`AUXiliary:OUTPut<m>:MEASure:OFFSet <p>`	Set offset as % of full scale (-999 to +999)
`AUXiliary:OUTPut<m>:MEASure:OFFSet?`	Query offset (%)

AUXiliary:OUTPut<m>:MEASure:EXPand <x>	Set expand multiplier (1, 2, 4, 8, 16, 32, 64, 128, 256)
AUXiliary:OUTPut<m>:MEASure:EXPand?	Query expand

CAPTure Subsystem

Command	Description
CAPTure:CONFig <c>	Set buffer channel configuration (X, XY, RT, XYRT)
CAPTure:CONFig?	Query config
CAPTure:LENgth <n>	Set capture buffer size (2–4096 kbytes)
CAPTure:LENgth?	Query buffer size (kbytes)
CAPTure:RATE <r>	Set sample rate in Hz
CAPTure:RATE?	Query rate (Hz)
CAPTure:MODE <m>	Set capture mode (ONEShot, LOOP, LOG)
CAPTure:MODE?	Query mode
CAPTure:TRIGger <b>	Enable/disable trigger-start on Ch-A Ref In rising edge (ON, OFF)
CAPTure:TRIGger?	Query trigger-start
CAPTure:STARt	Arm and start capture
CAPTure:STOP	Stop capture
CAPTure:STATus?	Returns IDLE, ARMED, RUNNING, or FULL
CAPTure:POINts?	Returns number of points currently stored
CAPTure:DATA? <first>,<count>	Read count samples starting at first (IEEE float binary blob)

STReam Subsystem

Command	Description
STReam:ENAble <b>	Start/stop Ethernet data streaming (ON, OFF)
STReam:ENAble?	Query stream state
STReam:CONFig <c>	Set streamed channel configuration (X, XY, RT, XYRT)
STReam:CONFig?	Query config
STReam:RATE <n>	Set stream rate divisor exponent (0–15)
STReam:RATE?	Query divisor

`STReam:FORMat <f>`	Set binary data format (`FLOAT32`, `INT16`)
`STReam:FORMat?`	Query format
`STReam:PACKet <b>`	Set packet size in bytes (128, 256, 512, 1024)
`STReam:PACKet?`	Query packet size
`STReam:PORT <p>`	Set UDP/TCP destination port (1024–65535)
`STReam:PORT?`	Query port
`STReam:ENDian <e>`	Set byte order (`LITTle`, `BIG`)
`STReam:ENDian?`	Query byte order

SCAN Subsystem

Command	Description
`SCAN:TARGet <t>`	Set scan target (`FREQ:A`, `FREQ:B`, `AMPL:A`, `AMPL:B`, `DC:A`, `DC:B`, `AUX1`, `AUX2`)
`SCAN:TARGet?`	Query target
`SCAN:STARt <v>`	Set sweep start value (full range of target parameter)
`SCAN:STARt?`	Query start
`SCAN:STOP <v>`	Set sweep stop value (full range of target parameter)
`SCAN:STOP?`	Query stop
`SCAN:SHAPe <s>`	Set sweep shape (`LINear`, `LOGarithmic`)
`SCAN:SHAPe?`	Query shape
`SCAN:DURation <t>`	Set sweep duration (0.001–99999 s)
`SCAN:DURation?`	Query duration
`SCAN:END <e>`	Set end behavior (`ONCE`, `REPeat`, `UPDOwn`)
`SCAN:END?`	Query end behavior
`SCAN:RUN`	Start scan
`SCAN:STOP:IMMediate`	Halt scan immediately
`SCAN:STATus?`	Returns `IDLE`, `RUNNING`, or `PAUSed`

FFT Subsystem

Command	Description
`FFT:SOURce <s>`	Set FFT input signal (`ADC`, `MIXer`, `FILTer`)
`FFT:SOURce?`	Query source

`FFT:CHANnel <n>`	Select channel for FFT (`A`, `B`)
`FFT:CHANnel?`	Query FFT channel
`FFT:AVERage <n>`	Set number of RMS averages (1, 3, 10, 30, 100)
`FFT:AVERage?`	Query averages
`FFT:SCALe <s>`	Set vertical scale in dB/div (1, 2, 5, 10, 20, 50, 100, 200)
`FFT:SCALe?`	Query scale
`FFT:REFLevel <r>`	Set top-of-screen reference level (-200 to +40 dBV)
`FFT:REFLevel?`	Query reference level
`FFT:DATA?`	Return 1024-point FFT (dBV values, comma-separated)

SYSTem Subsystem

Command	Description
`SYSTem:LINE <f>`	Set mains frequency for notch auto-tuning (`50`, `60`)
`SYSTem:LINE?`	Query mains frequency
`SYSTem:TIMEbase <s>`	Set 10 MHz timebase source (`INTernal`, `EXTernal`)
`SYSTem:TIMEbase?`	Query timebase source
`SYSTem:ETHernet:MODE <m>`	Set IP mode (`DHCP`, `STATic`)
`SYSTem:ETHernet:MODE?`	Query IP mode
`SYSTem:ETHernet:IP <addr>`	Set static IP address (IPv4 dotted-quad; takes effect on next boot)
`SYSTem:ETHernet:IP?`	Query current IP address
`SYSTem:ETHernet:MAC?`	Query MAC address (read-only)
`SYSTem:KCLick <b>`	Enable/disable key click (`ON`, `OFF`)
`SYSTem:ALARm <b>`	Enable/disable overload alarm (`ON`, `OFF`)
`SYSTem:ERRor:CLEar`	Clear ERR LED and error queue
`SYSTem:VERSion?`	Return firmware version string
`SYSTem:TRIP:CLEar`	Clear latched TRIP condition (hardware must be safe)
`SYSTem:LOCal`	Return to local (front panel enabled)

`SYSTem:REMote`	Set remote mode (front panel locked out)
`SYSTem:RWLock`	Remote with local lockout (no front-panel return)

STATus Subsystem

Command	Description
`STATus:OPERation?`	Query Operation Status register
`STATus:OPERation:ENAble <mask>`	Set Operation Status Enable register
`STATus:OPERation:ENAble?`	Query Operation Status Enable register
`STATus:QUEStionable?`	Query Questionable Status register
`STATus:QUEStionable:ENAble <mask>`	Set Questionable Status Enable register
`STATus:QUEStionable:ENAble?`	Query Questionable Status Enable register
`STATus:LIA?`	Query lock-in status word (overload, unlock, etc.)
`STATus:LIA:ENAble <mask>`	Set lock-in status enable register
`STATus:LIA:ENAble?`	Query lock-in status enable register
`STATus:ERRor?`	Query error status byte
`STATus:ERRor:ENAble <mask>`	Set error status enable register
`STATus:ERRor:ENAble?`	Query error status enable register
`STATus:PRESet`	Reset all enable registers to default values

DISPlay Subsystem

Command	Description
`DISPlay:LAYout <l>`	Set screen layout (`FULL`, `DUAL`, `QUAD`, `CHART`, `FFT`, `SCAN`)
`DISPlay:LAYout?`	Query layout
`DISPlay:CHANnel<n>:QUANtity <q>`	Set data pane n quantity (`X`, `Y`, `R`, `THETa`, `XNOise`, `YNOise`, `FREQuency`)
`DISPlay:CHANnel<n>:QUANtity?`	Query data pane n quantity
`DISPlay:BLANk <b>`	Blank the display (`ON`, `OFF`)
`DISPlay:BLANk?`	Query blanking state
`DISPlay:CAPTure`	Capture a screen image to USB storage

1 Getting Started

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The measurements in this chapter are designed to acquaint first-time users with the SR835 Dual Lock-In Amplifier. Do not be concerned if your results do not agree exactly with the examples —the focus is on learning how to operate the instrument, not on achieving a specific numerical result. Work through these exercises before attempting real experiments.

1.1 Notation

`[Key]` A front-panel key or button.

`{Touch}` A touchscreen button, tile, or icon.

`[↑] / [↓]` The increment and decrement buttons. When a parameter tile is selected on the touchscreen, these buttons step its value up or down in fine increments. Touch and hold a tile to open a numeric keypad for direct entry.

1.2 SR835 Front Panel

The SR835 provides two independent measurement channels, **Ch 1** and **Ch 2**, each with its own signal input, phase-sensitive detector, cascaded low-pass filter, and analog output. Both channels share a single numerically controlled oscillator whose frequency, amplitude, and phase are configured from the Reference section of the front panel.

Most settings are adjusted through the touchscreen. Touch a parameter tile to select it; the tile highlights and the [↑]/[↓] buttons then step through its values. Touch and hold to open a numeric keypad. The info bar across the top of the screen always shows the current reference phase, frequency, detect harmonic, and sine amplitude for each channel.

1.2.1 Signal Path

The input range sets the largest signal a channel can accept before overload. Reduce the range as far as possible without triggering an overload indicator; this maximises ADC utilisation and lowers the noise floor. The sensitivity scales the analog output and numeric displays but does not affect the measured value of X, Y, R, or θ—it only determines what signal level corresponds to full-scale output.

1.2.2 Synchronous Filter

The SR835 includes an always-active synchronous comb filter that suppresses harmonics of the reference in the demodulated output.

Unlike earlier lock-ins that required enabling this filter only at low frequencies, the SR835 applies it continuously across its full 1 mHz–1 MHz range with no user action required.

1.3 The Basic Lock-In

This exercise uses the internal oscillator to explore basic lock-in operation on Ch 1. You will need a BNC cable.

Do This	Explanation
1. Disconnect all cables. Power on the SR835 while holding [Reset].	Holding [Reset] at power-on restores all standard default settings. See the Standard Settings table in the Operation chapter for the complete list.
2. Connect Sine Out to Ch 1 Input (A) using a BNC cable.	The lock-in defaults to internal oscillator mode at 1.000 kHz. The Sine Out amplitude defaults to 0 Vrms, so X, Y, and R all read zero; θ is noise.
3. Touch {Ampl} in the info bar. Enter {5}{0}{0}{mV} on the keypad.	With 500 mVrms output and near-zero phase shift, Ch 1 X should read approximately 0.5000 V and Y should read near zero.
4. Press [Auto Phase].	Automatically adjusts the reference phase to null any residual phase error. Y and θ should go to zero.
5. Press [+90°].	Adds 90° to the reference phase. X drops to zero and Y becomes −0.5000 V. Touch {Phase} in the info bar and use [↑]/[↓] to return the phase to zero, or press [Auto Phase] again as a shortcut.
6. Touch {Fint} in the info bar. Enter {1}{0}{kHz}. Then use [↑]/[↓] to fine-tune the frequency.	Direct entry changes the frequency immediately. X and R should remain within 1 % of 500 mV at all frequencies. The [↑]/[↓] buttons are convenient for small adjustments.
7. Touch {Ampl} and use [↑]/[↓] to reduce	X and R follow the amplitude change. The input overload indicator

the sine amplitude to 5.0 mV.	may light, signalling that a lower input range is available.
8. Press [Auto Range] for Ch 1.	Adjusts the Ch 1 input range to the highest gain setting that avoids overload. The range should step to 10 mV and the signal-strength indicator should rise.
9. Press [Time Constant ↓] repeatedly to select 300 µs.	On most lock-ins this would cause severe noise from the unrejected 2f component. On the SR835, the always-active synchronous filter suppresses 2f continuously, so the outputs remain much cleaner than on an older instrument at the same short time constant.
10. Press [Filter Slope] to cycle through 6, 12, 18, and 24 dB/oct.	Additional filter poles reduce residual out-of-band noise further. The synchronous filter benefit is most visible at 6 dB/oct.
Return to 6 dB/oct when done.	
This concludes the basic lock-in exercise. You should now be comfortable with the front-panel workflow: setting the oscillator, adjusting the input range, reading X/Y/R/θ, and observing the effect of the synchronous filter.	

1.4 Dual-Channel Measurement

This exercise uses both measurement channels simultaneously to demonstrate independent harmonic detection on a shared oscillator. You will need two BNC cables and a BNC tee.

Do This	Explanation

1. Reset the instrument as in the previous exercise. Using a BNC tee, connect Sine Out to both Ch 1 Input (A) and Ch 2 Input (B).	Both channels measure the same source signal. The oscillator is shared, so both demodulators see the same reference frequency and phase.
2. Touch {Ampl} and set the sine amplitude to 500 mV.	With default harmonic settings (both channels at 1×), Ch 1 X and Ch 2 X should each read approximately 0.5000 V.
3. On the Ch 2 channel panel, touch the harmonic tile and enter {2}.	Ch 2 now detects at the second harmonic (2 kHz). Because the Sine Out is a clean sinusoid there is no signal at 2f, so Ch 2 X, Y, and R drop to near zero. Ch 1 is unaffected.
4. Press [Auto Phase] for Ch 1.	Each channel's reference phase is adjusted independently. Phasing Ch 1 has no effect on Ch 2.
5. Touch {Fint} and set the frequency to 500 Hz. Observe both channels.	Ch 1 detects at 500 Hz (fundamental); Ch 2 at 1 kHz (second harmonic). Each channel independently measures its assigned spectral component from the same input. This is the principal advantage of dual-channel operation—no second instrument, no synchronisation cables.
6. Return Ch 2 harmonic to {1}. Verify that both channels read ≈500 mV.	With both channels at the fundamental, any remaining difference reflects cable path mismatch rather than instrument error.

1.5 Using Displays

This exercise explores the strip chart, FFT, and other display modes. You will need a synthesised function generator capable of 500 mVrms at 1.000 kHz, BNC cables, and an appropriate terminator.

Do This	Explanation
1. Reset. Connect the function generator (1.000 kHz, 500 mVrms) to Ch 1 Input (A).	The generator output impedance is typically 50 Ω or 600 Ω. Use a feedthrough or T-terminator as appropriate; without one, the displayed amplitude will not agree with the generator setting. The lock-in input impedance is 10 MΩ.
2. Touch {Fint} and set the internal frequency to 999.800 Hz.	Setting the lock-in 0.2 Hz below the signal frequency makes X and Y oscillate at 0.2 Hz (the difference frequency). R is phase-independent and remains near 500 mV. The slowly rotating θ is easier to see in the strip chart than as a single number.
3. Press [Screen Layout] once to switch to the full-screen strip chart.	New data plots at the right edge; older data scrolls left. The scroll rate is set by the horizontal time scale. The info bar remains at the top; touch a tile to change a parameter.
4. Touch the orange {θ} scale tile at the bottom. Touch {Autoscale}. Touch the tile again to dismiss the palette.	The vertical scale palette adjusts how one data channel is displayed. {Autoscale} sets scale and centre to best fill the screen. θ ramps ±180° so the autoscaled range is about 50°/div.
5. Auto-scale the X (green), Y (blue), and R (yellow) channels in turn. For R, touch {Center Newest Point} then zoom in several steps with {Zoom In}.	X and Y are 500 mV sine waves at 0.2 Hz. R is a flat line near 500 mV; zooming in reveals the noise floor. The small triangle on the right edge marks zero for each channel—after zooming, the R zero triangle moves off the graph edge.
6. Touch the white {Time} scale tile. Use	Horizontal scale changes apply to all channels simultaneously. The chart

{Zoom Out} to show more history.	always anchors the most recent point at the right edge.
7. Touch {Pause}. Move the cursor with [↑]/[↓] or by dragging to inspect historical values.	The cursor reads the min, max, or mean of the data within the cursor width at its position. The time tile at the left of the scale bar shows elapsed time; touch it to toggle to absolute time and date. Data continues accumulating in the background while the chart is paused. Touch {Resume} to return to live scrolling.
8. Increase the generator amplitude to 1.5 Vrms. Observe the chart. Then return to 500 mVrms.	The signal exceeds the 1 V input range. Overload indicators appear on the affected channels and a violet band marks the overloaded region in the chart history, so the data quality can be assessed after the fact.
9. Press [Screen Layout] repeatedly to step through the half-screen strip chart, full-screen FFT, half-screen FFT, and full numeric display. Press [Screen Layout] again to return to the trend graph.	The FFT shows the ADC or demodulator spectrum. Use the {Src}, {dB}, {Hz}, {Avgs}, and {Live}/{Paused} tiles to control it. The full numeric screen adds Aux Input readouts and Aux Output control tiles.

1.6 Sensitivity, Offset and Expand

This exercise explores output scaling. You will need a BNC cable and a digital voltmeter (DVM).

Do This	Explanation
1. Reset. Connect Sine Out to Ch 1 Input (A). Connect the Ch 1 Analog	The Ch 1 analog output voltage is $\mathrm{CH1} = \left(\frac{X}{S_{\mathrm{ens}}} - \mathrm{Offset}\right) \times \mathrm{Expand} \times 10\ \mathrm{V}$ where S_{ens} is the sensitivity and Offset is expressed as a fraction.

Output to the DVM (DC volts).	
2. Touch {Ampl} and set the sine to 100 mV.	Default: sensitivity 1 V, offset 0 %, expand ×1. Output = (0.1/1.0)×1×10 V = 1 V. The bar graph shows 10 %.
3. Press [Sensitivity ↓] to 500 mV, then to 100 mV.	At 500 mV sensitivity: output = 2 V (20 % full scale). At 100 mV: output = 10 V (full scale). The strip-chart trace of X is unchanged—sensitivity only rescales the outputs and displays, not the measurement.
4. Touch {Offset} for Ch 1. Touch {Auto}.	Auto Offset adjusts X until the displayed value reads zero. The offset is expressed as a percentage of the sensitivity (here ≈100 %). The DVM goes to 0 V. An "Ofst" indicator appears on the Ch 1 display. Note that the strip-chart trace of X shifts—offset modifies the actual value of X, unlike sensitivity.
5. Enter an offset of 90 %.	Output = (0.1/0.1 − 0.9)×1×10 V = 1 V. The bar graph is at 10 %.
6. Press [Expand ↑] to select ×10.	Output = (0.1/0.1 − 0.9)×10×10 V = 10 V. The bar graph is at 100 % and the numeric readout gains one digit of resolution. An "Expd" indicator appears. Expand does not change the value of X; the strip-chart trace is unaffected.
7. Press [Expand ↑] to select ×100.	The output overload LED lights. The output would need to reach 100 V but is clamped. The strip-chart trace and numeric readout remain correct.
8. Press [Expand] to disable expand. Touch {Offset} to disable the offset.	The DVM returns to 10 V. With offset and expand, the analog output gain and bias can be configured for feedback control or signal conditioning

without altering the underlying measurement.

1.7 Saving and Recalling Setups

The SR835 stores up to 8 complete instrument setups in non-volatile memory.

Do This	Explanation
1. Press and hold [System] to open the system menu.	The system menu configures instrument-level settings: clock, interfaces, and file numbering. Set the time and date here so that saved setups are labelled correctly.
2. Touch {h} to highlight the hours field; use the keypad to enter the hour in 24-hour format. Set minutes and seconds similarly. Touch {Time set} to commit. Set the date in the same way.	The time and date stamp each saved setup, making it easier to identify the correct one when recalling later.
3. Close the system menu. Set the sensitivity to 100 mV, the time constant to 1 s, and the filter slope to 12 dB/oct.	A non-default configuration is needed to demonstrate save and recall.
4. Press [Save/Recall]. Touch {Save} next to location 1. Enter a name on the keypad. Touch {Confirm}.	Setups 1 through 8 are available. Each tile displays the setup name and the time and date it was saved.
5. Change the sensitivity, time constant, and filter slope to different	A summary of all settings that will change is shown before confirming, allowing the recall to be cancelled if necessary. After confirm, the sen-

values. Press [Save/Recall] again. Touch {Recall} for location 1, review the summary of changes, then touch {Confirm}.	sitivity, time constant, and slope return to the saved values.

1.8 Aux Outputs and Inputs

The SR835 has four programmable auxiliary DC outputs and four analog inputs on the rear panel. You will need BNC cables and a DVM.

Do This	Explanation
1. Reset. Connect Aux Out 1 (rear panel) to the DVM.	The four Aux Outputs provide programmable voltages from −10.5 V to +10.5 V. They can be set from the front panel or via the remote interface, and are useful for controlling experiment parameters such as temperature, pressure, or position.
2. Press [Aux Output]. Touch {1}{0}{Enter} to set Aux Out 1 to 10.000 V.	The DVM should follow. Increment buttons provide fine resolution; the keypad is faster for large changes.
Then use [↑]/[↓] to adjust to 5.000 V.	
3. Press [Screen Layout] to the full numeric display.	This screen shows all four Aux Input readings alongside the lock-in data channels. The four Aux Output values are shown in tiles across the bottom; touch a tile to open its keypad directly.
4. Disconnect the DVM. Connect Aux Out 1 to Aux In 1 on the rear panel.	Aux In 1 should now read 5.000 V. The Aux Inputs accept ±10 V and can be assigned to a strip-chart data channel using [Config], so they can

be logged and plotted alongside lock-in outputs.

1.9 Scanning

This exercise uses the scan engine to sweep the internal reference frequency through an external signal. You will need a synthesised function generator (500 mVrms at 1.000 kHz) and BNC cables.

Do This	Explanation
1. Reset. Connect the function generator (1.000 kHz, 500 mVrms) to Ch 1 Input (A). Use an appropriate terminator.	X and Y display slowly changing values because the lock-in and generator are at nearly but not exactly the same frequency. R is steady near 500 mV.
2. Press and hold [Scan] to open the scan menu.	Scannable parameters include Finternal, Amplitude, DC Level, and Aux Outputs 1–4. Scans can be linear or logarithmic, and can run once, repeat continuously, or reverse at each end. The Scan Duration sets the total time from Begin to End Value; the Parameter Update Interval sets the step size. Shorter update intervals produce smoother scans; longer ones allow the experiment more time to settle at each step.
3. Leave the scan parameter at Finternal. Set Begin Value to 990 Hz, End Value to 1010 Hz, and End Mode to Repeat. Touch {Close}.	A 20 Hz span centred on the signal frequency. The scan is configured but not yet running; the oscillator has not changed.
4. Press [Scan] briefly to arm. Press [Play/Pause] briefly to start.	The Ready LED lights when armed, and the frequency shown in the info bar changes to the Begin Value displayed in orange. The Run LED

	lights when scanning. The frequency increases until it reaches the End Value, then in Repeat mode resets and runs again.
5. Press [Screen Layout] to the full-screen strip chart. Touch the white {Time} scale tile and zoom out to 10 S/div.	Strip charts are much easier than trend graphs for visualising a scan. A 20 s scan fits in two divisions at 10 S/div.
6. Dismiss the X and Y traces. Autoscale R (yellow). Touch the orange {θ} tile and set the scale to 200°/div.	R shows the lock-in magnitude response as the frequency sweeps through 1 kHz. θ shows the phase response.
7. Press [Config]. Assign the blue data channel to Finternal. Touch {Close}. Autoscale the blue trace.	The frequency ramp is now overlaid in the chart history alongside R and θ, confirming the timing of the sweep relative to the lock-in response.
8. Press and hold [Scan] to return to the scan menu. Change End Mode to Once. Touch {Close}. Press [Scan] briefly to stop, then again to re-arm. Press [Play/Pause] to start the new scan.	In Once mode the scan stops at the End Value and the strip chart pauses automatically. The Done LED lights. Use the cursor to measure the peak of R and read the frequency at which the resonance occurs.
9. Touch {Resume} in the horizontal scale palette to restart live scrolling.	Data collection continued while the chart was paused. Resuming realigns the display so the current point is back at the right edge.
10. Press and hold [Play/Pause] to reset the	Disarming returns the internal frequency to its pre-scan value, shown

scan to the Begin Value.

Press [Scan] briefly to disarm.

in white in the info bar. The Done LED extinguishes.

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2.1 What is a Lock-in Amplifier?

Lock-in amplifiers are used to detect and measure very small ac signals—all the way down to a few nanovolts. Accurate measurements may be made even when the small signal is obscured by noise sources many thousands of times larger.

Lock-in amplifiers use a technique known as *phase-sensitive detection* to single out the component of the signal at a specific reference frequency and phase. Noise signals at frequencies other than the reference frequency are rejected and do not affect the measurement.

2.1.1 Why Use a Lock-in?

Consider an example. Suppose the signal is a 10 nV sine wave at 10 kHz. Clearly some amplification is required. A good low-noise amplifier will have about 5 nV/$\sqrt{\text{Hz}}$ of input noise. If the amplifier bandwidth is 100 kHz and the gain is 1000, then the output will be 10 μV of signal (10 nV × 1000) and 1.6 mV of broadband noise (5 nV/$\sqrt{\text{Hz}}$ × $\sqrt{100}$ kHz × 1000). We will not have much luck measuring the output signal unless we single out the frequency of interest.

If we follow the amplifier with a bandpass filter with a Q = 100 (a very good filter) centred at 10 kHz, any signal in a 100 Hz bandwidth will be detected (10 kHz/Q). The noise in the filter passband will be 50 μV (5 nV/$\sqrt{\text{Hz}}$ × $\sqrt{100}$ Hz × 1000) and the signal will still be 10 μV. The output noise is still much greater than the signal and an accurate measurement cannot be made. Further gain will not help the signal-to-noise problem.

Now try following the amplifier with a phase-sensitive detector (PSD). The PSD can detect the signal at 10 kHz with a bandwidth as narrow as 0.01 Hz! In this case, the noise in the detection bandwidth will be only 0.5 μV (5 nV/$\sqrt{\text{Hz}}$ × $\sqrt{0.01}$ Hz × 1000) while the signal is still 10 μV. The signal-to-noise ratio is now 20 and an accurate measurement of the signal is possible.

2.1.2 What is Phase-Sensitive Detection?

Lock-in measurements require a frequency reference. Typically an experiment is excited at a fixed frequency (from an oscillator or function generator) and the lock-in detects the response from the experiment at the reference frequency. In the diagram below, the reference signal is a square wave at frequency f_{ref}. This might

be the sync output from a function generator. If the sine output from the function generator is used to excite the experiment, the response might be the signal waveform shown below. The signal is $V_{\text{sig}} \sin(\omega_r t + \theta_{\text{sig}})$ where $\omega_r = 2\pi f_{\text{ref}}$ and V_{sig} is the signal amplitude.

The SR835 generates its own sine wave at frequency f_L, shown as the lock-in reference below. The lock-in reference is $\sin(\omega_L t + \theta_{\text{ref}})$ where $\omega_L = 2\pi f_L$.

The SR835 amplifies the signal and then multiplies it by the lock-in reference using a phase-sensitive detector (PSD) or multiplier. The output of the PSD is simply the product of two sine waves.

$$\begin{aligned} V_{\text{psd}} &= V_{\text{sig}} \sin(\omega_r t + \theta_{\text{sig}}) \sin(\omega_L t + \theta_{\text{ref}}) \\ &= \frac{1}{2} V_{\text{sig}} \cos([\omega_r - \omega_L]t + \theta_{\text{sig}} - \theta_{\text{ref}}) \\ &\quad - \frac{1}{2} V_{\text{sig}} \cos([\omega_r + \omega_L]t + \theta_{\text{sig}} + \theta_{\text{ref}}) \end{aligned}$$

The PSD output is two ac signals, one at the difference frequency $(\omega_r - \omega_L)$ and the other at the sum frequency $(\omega_r + \omega_L)$.

If the PSD output is passed through a low-pass filter, the ac signals are removed. What will be left? In the general case, nothing. However, if ω_r equals ω_L, the difference frequency component will be a dc signal. In this case, the filtered PSD output will be

$$V_{\text{psd}} = \frac{1}{2} V_{\text{sig}} \cos(\theta_{\text{sig}} - \theta_{\text{ref}})$$

This is a very nice signal—it is a dc signal proportional to the signal amplitude.

2.1.3 Narrow-Band Detection

Now suppose the input is made up of signal plus noise. The PSD and low-pass filter only detect signals whose frequencies are very close to the lock-in reference frequency. Noise signals at frequencies far from the reference are attenuated at the PSD output by the low-pass filter (neither $\omega_{\text{noise}} - \omega_{\text{ref}}$ nor $\omega_{\text{noise}} + \omega_{\text{ref}}$ are close to dc). Noise at frequencies very close to the reference frequency will result in very low frequency ac outputs from the PSD ($|\omega_{\text{noise}} - \omega_{\text{ref}}|$ is small). Their attenuation depends upon the low-pass filter bandwidth and roll-off. A narrower bandwidth will remove noise sources very close to the reference frequency; a wider bandwidth allows these signals

to pass. The low-pass filter bandwidth determines the bandwidth of detection. Only the signal at the reference frequency will result in a true dc output and be unaffected by the low-pass filter. This is the signal we want to measure.

2.1.4 Where Does the Lock-in Reference Come From?

We need to make the lock-in reference the same as the signal frequency, i.e. $\omega_r = \omega_L$. Not only do the frequencies have to be the same, the phase between the signals cannot change with time, otherwise $\cos(\theta_{\text{sig}} - \theta_{\text{ref}})$ will change and V_{psd} will not be a dc signal. In other words, the lock-in reference needs to be *phase-locked* to the signal reference.

Lock-in amplifiers use a phase-locked loop (PLL) to generate the reference signal. An external reference signal (in this case, the reference square wave) is provided to the lock-in. The PLL in the lock-in locks the internal reference oscillator to this external reference, resulting in a reference sine wave at ω_r with a fixed phase shift of θ_{ref}. Since the PLL actively tracks the external reference, changes in the external reference frequency do not affect the measurement.

> *All lock-in measurements require a reference signal.*

In this case, the reference is provided by the excitation source (the function generator). This is called an *external reference* source. In many situations, the SR835's internal oscillator may be used instead. The internal oscillator is a function generator (with variable sine output and a TTL sync) which is always phase-locked to the reference oscillator.

2.1.5 Magnitude and Phase

Remember that the PSD output is proportional to $V_{\text{sig}} \cos\theta$ where $\theta = (\theta_{\text{sig}} - \theta_{\text{ref}})$. θ is the phase difference between the signal and the lock-in reference oscillator. By adjusting θ_{ref} we can make θ equal to zero, in which case we can measure V_{sig} $(\cos\theta = 1)$. Conversely, if θ is 90°, there will be no output at all. A lock-in with a single PSD is called a single-phase lock-in and its output is $V_{\text{sig}} \cos\theta$.

This phase dependency can be eliminated by adding a second PSD. If the second PSD multiplies the signal with the reference oscillator shifted by 90°, i.e. $\sin(\omega_L t + \theta_{\text{ref}} + 90°)$, its low-pass filtered output will be

$$V_{\text{psd2}} = \frac{1}{2} V_{\text{sig}} \sin\left(\theta_{\text{sig}} - \theta_{\text{ref}}\right) \approx V_{\text{sig}} \sin\theta$$

Now we have two outputs, one proportional to $\cos\theta$ and the other proportional to $\sin\theta$. If we call the first output X and the second Y,

$$X = V_{\text{sig}} \cos\theta \qquad Y = V_{\text{sig}} \sin\theta$$

these two quantities represent the signal as a vector relative to the lock-in reference oscillator. X is called the *in-phase* component and Y the *quadrature* component. This is because when $\theta = 0$, X measures the signal while Y is zero.

By computing the magnitude (R) of the signal vector, the phase dependency is removed.

$$R = \left(X^2 + Y^2\right)^{1/2} = V_{\text{sig}}$$

R measures the signal amplitude and does not depend upon the phase between the signal and lock-in reference.

A dual-phase lock-in, such as the SR835, has two PSDs with reference oscillators 90° apart, and can measure X, Y, and R directly. In addition, the phase θ between the signal and lock-in reference can be measured according to

$$\theta = \arctan(Y/X)$$

2.2 What Does a Lock-in Measure?

So what exactly does the SR835 measure? Fourier's theorem basically states that any input signal can be represented as the sum of many sine waves of differing amplitudes, frequencies, and phases. This is generally considered as representing the signal in the "frequency domain". Normal oscilloscopes display the signal in the "time domain". Except in the case of clean sine waves, the time-domain representation does not convey very much information about the various frequencies that make up the signal.

The SR835 multiplies the input signal by a pure sine wave at the reference frequency. All components of the input signal are multiplied by the reference simultaneously. Mathematically speaking, sine waves of differing frequencies are orthogonal, i.e. the average of the product of two sine waves is zero unless the frequencies are exactly the same. In the SR835, the product of this multiplication

yields a dc output signal proportional to the component of the signal whose frequency is exactly locked to the reference frequency. The low-pass filter which follows the multiplier provides the averaging which removes the products of the reference with components at all other frequencies.

The SR835, because it multiplies the signal with a pure sine wave, measures the single Fourier (sine) component of the signal at the reference frequency. Let's take a look at an example. Suppose the input signal is a simple square wave at frequency f. The square wave is actually composed of many sine waves at multiples of f with carefully related amplitudes and phases. A 2 V pk–pk square wave can be expressed as

$$S(t) = 1.273\sin(\omega t) + 0.4244\sin(3\omega t) + 0.2546\sin(5\omega t) + ...$$

where $\omega = 2\pi f$. The SR835, locked to f, will single out the first component. The measured signal will be $1.273\sin(\omega t)$, not the 2 V pk–pk that you would measure on an oscilloscope.

In the general case, the input consists of signal plus noise. Noise is represented as varying signals at all frequencies. The ideal lock-in only responds to noise at the reference frequency. Noise at other frequencies is removed by the low-pass filter following the multiplier. This "bandwidth narrowing" is the primary advantage that a lock-in amplifier provides. Only inputs at the reference frequency result in an output.

2.2.1 RMS or Peak?

Lock-in amplifiers as a general rule display the input signal in volts rms. When the SR835 displays a magnitude of 1 V (rms), the component of the input signal at the reference frequency is a sine wave with an amplitude of 1 Vrms or 2.8 V pk–pk.

Thus, in the previous example with a 2 V pk–pk square wave input, the SR835 would detect the first sine component, $1.273\sin(\omega t)$. The measured and displayed magnitude would be 0.90 V (rms), i.e. $1.273/\sqrt{2}$.

2.2.2 Degrees or Radians?

In this discussion, frequencies have been referred to as f (Hz) and ω ($2\pi f$ radians/sec). This is because people measure frequencies in cycles per second and math works best in radians. For purposes

of measurement, frequencies as measured in a lock-in amplifier are in Hz. The equations used to explain the actual calculations are sometimes written using ω to simplify the expressions.

Phase is always reported in degrees. Once again, this is by custom. Equations written as $\sin(\omega t + \theta)$ are written as if θ is in radians mostly for simplicity. Lock-in amplifiers always manipulate and measure phase in degrees.

2.3 The Reference Oscillator

A lock-in amplifier requires a reference oscillator phase-locked to the signal frequency. In general, this is accomplished by phase-locking an internal oscillator to an externally provided reference signal. This reference signal usually comes from the signal source that is providing the excitation to the experiment.

2.3.1 Reference Input

The SR835 reference input can trigger on an analog signal (like a sine wave) or a TTL logic signal. The first case is called External Sine. The input is ac coupled (above 1 Hz) and the input impedance is 1 MΩ. A sine wave input greater than 200 mV pk is required. Positive zero crossings are detected and considered to be the zero for the reference phase shift.

TTL reference signals can be used at all frequencies up to 1 MHz. For frequencies below 1 Hz, a TTL reference signal is required. Many function generators provide a TTL SYNC output which can be used as the reference. This is convenient since the generator's sine output might be smaller than 200 mV or be varied in amplitude. The SYNC signal will provide a stable reference regardless of the sine amplitude.

When using a TTL reference, the reference input trigger can be set to Pos Edge (detect rising edges) or Neg Edge (detect falling edges). In each case, the internal oscillator is locked (at zero phase) to the detected edge.

2.3.2 Internal Oscillator

The internal oscillator in the SR835 is a 1 MHz function generator with sine and sync outputs. The oscillator generates a digitally synthesised sine wave. The internal oscillator sine wave is output at the Sine Out BNC on the front panel. The amplitude of this output may be set from 1 nV to 2 V (rms).

When an external reference is used, this internal oscillator sine wave is phase-locked to the reference. The rising zero crossing is locked to the detected reference zero crossing or edge. In this mode, the Sine Out provides a sine wave phase-locked to the external reference. Phase locking is accomplished digitally.

The internal oscillator may be used without an external reference. In the Internal Reference mode the frequency is set in the lock-in and the Sine Out provides the excitation for the experiment. The phase-locked loop is not used in this mode.

The TTL Out on the rear panel provides a TTL sync output. The internal oscillator's rising zero crossings are detected and translated to TTL levels. This output is a square wave.

2.3.3 Reference Oscillators and Phase

The internal oscillator sine wave is not the reference signal to the phase-sensitive detectors. The SR835 computes a second sine wave, phase shifted by θ_{ref} from the internal oscillator (and thus from an external reference), as the reference input to the X phase-sensitive detector. This waveform is $\sin(\omega_r t + \theta_{\text{ref}})$. The reference phase shift is adjustable in 0.001° increments.

The input to the Y PSD is a third sine wave, computed by the SR835, shifted by 90° from the second sine wave. This waveform is $\sin(\omega_r t + \theta_{\text{ref}} + 90°)$.

The phase shifts (θ_{ref} and the 90° shift) are exact numbers and accurate to better than 0.001°. Neither waveform is actually output in analog form since the phase-sensitive detectors are digital multipliers inside the SR835.

2.3.4 Phase Jitter

When an external reference is used, the phase-locked loop contributes a small amount of phase jitter. The internal oscillator is supposed to be locked with zero phase shift relative to the external reference. Phase jitter means that the average phase shift is zero but the instantaneous phase shift has a few millidegrees of noise. This shows up at the output as noise in phase or quadrature measurements.

Phase noise can also cause noise to appear at the X and Y outputs. This is because a reference oscillator with significant phase noise is the same as a reference whose frequency spectrum is spread out.

That is, the reference is not a single frequency, but a distribution of frequencies about the true reference frequency. These spurious frequencies are attenuated quite a bit but can still cause problems. The spurious reference frequencies result in signals close to the reference being detected. Noise at nearby frequencies now appears near dc and affects the lock-in output.

Phase noise in the SR835 is very low and generally causes no problems. In applications requiring no phase jitter, the internal reference mode should be used. Since there is no PLL, the internal oscillator and the reference sine waves are directly linked and there is no jitter in the measured phase. (Actually, the phase jitter is the phase noise of a crystal oscillator and is very, very small.)

2.3.5 Harmonic Detection

It is possible to compute the two PSD reference sine waves at a multiple of the internal oscillator frequency. In this case, the lock-in detects signals at $N \times f_{\mathrm{ref}}$ which are synchronous with the reference. The Sine Out frequency is not affected. The SR835 can detect at any harmonic up to N = 99 as long as $N \times f_{\mathrm{ref}}$ does not exceed 1 MHz.

2.4 The Phase-Sensitive Detectors

The amplified input signal is converted to digital form using an A/D converter. The SR835 then multiplies the signal with the reference sine waves digitally.

The digital PSDs in the SR835 act as linear multipliers—they multiply the signal by a reference sine wave. This approach eliminates many problems associated with analog PSDs. In particular:

- **Harmonic rejection.** Because the reference sine waves are computed digitally, they have very low harmonic content, meaning that only the signal at the reference frequency is detected. In contrast, a square-wave multiplying lock-in detects at all odd harmonics of the reference.
- **Output offsets.** The digital multiplication produces no erroneous dc output offsets. In an analog PSD, dc output offset contributes to error and zero drift.
- **Dynamic reserve.** The dynamic reserve of an analog PSD is limited to about 60 dB. In the digital lock-in, the dynamic reserve

is limited by the quality of the A/D conversion. Once the input signal is digitised, no further errors are introduced. The accuracy of the multiplication does not depend on the size of the numbers. The A/D converter used in the SR835 is extremely linear, meaning that the presence of large noise signals does not impair its ability to correctly digitise a small signal. The dynamic reserve of the SR835 can exceed 120 dB without any problems.

- **Gain stability.** In an analog linear PSD, any amplitude variation in the reference sine wave shows up directly as a variation in the overall gain. The digital reference sine wave has a precise amplitude and never changes, eliminating this source of gain error.

2.5 Time Constants and Sensitivity

Remember, the output of the PSD contains many signals. Most of the output signals have frequencies which are either the sum or difference between an input signal frequency and the reference frequency. Only the component of the input signal whose frequency is exactly equal to the reference frequency will result in a dc output.

The low-pass filter at the PSD output removes all of the unwanted ac signals, both the 2f (sum of the signal and the reference) and the noise components. This filter is what makes the lock-in such a narrow-band detector.

2.5.1 RC Filters

Traditionally, the time-constant setting of a lock-in amplifier determines the bandwidth of an RC low-pass filter. The time constant is simply $1/(2\pi f)$ where f is the -3 dB frequency of the filter. The low-pass filters are simple 6 dB/octave roll-off, RC-type filters. A 1-second time constant refers to a filter whose -3 dB point occurs at 0.16 Hz and rolls off at 6 dB/octave beyond 0.16 Hz. Typically, there are two successive filters so that the overall filter can roll off at either 6 dB or 12 dB per octave. The time constant refers to the -3 dB point of each filter alone, not the combined filter.

The notion of time constant arises from the fact that the actual output is supposed to be a dc signal. In fact, when there is noise at the input, there is noise on the output. By increasing the time constant, the output becomes more steady and easier to measure reliably. The trade-off comes when real changes in the input signal take many time constants to be reflected at the output. This is because a single

RC filter requires about 5 time constants to settle to its final value. The time constant reflects how slowly the output responds, and thus the degree of output smoothing.

The time constant also determines the equivalent noise bandwidth (ENBW) for noise measurements. The ENBW is not the filter −3 dB pole; it is the effective bandwidth for Gaussian noise. More about this later.

The digital signal processing in the SR835 handles all of the low-pass filtering. Each PSD can be followed by up to four filter stages for up to 24 dB/oct of roll-off.

Why are multiple filter stages desirable? Consider an example where the reference is at 1 kHz and a large noise signal is at 1.05 kHz. The PSD noise outputs are at 50 Hz (difference) and 2.05 kHz (sum). Clearly the 50 Hz component is the more difficult to low-pass filter. If the noise signal is 80 dB above the full-scale signal and we would like to measure the signal to 1% (−40 dB), then the 50 Hz component needs to be reduced by 120 dB. To do this in two stages would require a time constant of at least 3 seconds. To accomplish the same attenuation in four stages only requires 100 ms of time constant. In the second case, the output will respond 30 times faster and the experiment will take less time.

2.5.2 Synchronous Filters

Even if the input signal has no noise, the PSD output always contains a component at 2f (sum frequency of signal and reference) whose amplitude equals or exceeds the desired dc output depending upon the phase. At low frequencies, the time constant required to attenuate the 2f component can be quite long. For example, at 1 Hz, the 2f output is at 2 Hz and to attenuate the 2 Hz by 60 dB in two stages requires a time constant of 3 seconds.

A synchronous filter operates totally differently. The PSD output is averaged over a complete cycle of the reference frequency. The result is that all components at multiples of the reference (2f included) are notched out completely. In the case of a clean signal, almost no additional filtering would be required. This is increasingly useful the lower the reference frequency. Imagine what the time constant would need to be at 0.001 Hz!

The SR835 includes an always-active synchronous comb filter that operates across the full 1 mHz–1 MHz detection range. Unlike earlier lock-in amplifiers that required the user to enable this filter —and only offered it at low frequencies—the SR835 applies synchronous filtering continuously with no user action required. The combination of the synchronous filter with the cascaded RC low-pass filters removes all multiples of the reference frequency and provides overall noise attenuation as well.

2.5.3 Sensitivity and Input Range

The Input Range sets the analog gain in the SR835. The settings reflect the largest signal at the input before the amplifier overloads. Setting the Input Range to a smaller value increases the analog gain. The signal-strength LEDs on the front panel indicate how much of the A/D range is being used. The general rule is to decrease the Input Range as much as possible without overloading the amplifier. This increases the A/D range utilised and optimises the output noise performance of the lock-in.

The digital signal processing in the SR835 uses floating-point math. The outputs reflect the actual analog signal gain preceding the analog-to-digital converter and are simply the signal input voltages at the input BNCs. All digital filtering is performed using floating-point math, and these values have no real limit in size and do not overload.

The Sensitivity sets what input signal level corresponds to full-scale (10 V) at the analog output BNC. The Sensitivity also sets the scale for the displayed bar graphs and numerical readouts of X, Y, and R. Note that this is a numerical output conversion. Output overloads do not affect the actual measurement results. They only indicate that the output value exceeds 100% of the chosen Sensitivity and the output BNC, the bar graph, and the displayed numerical readout will be pinned at their maximums. The results displayed on the strip charts or available over the computer interfaces are the floating-point outputs and are unaffected by output overloads.

The Sensitivity is chosen to conveniently and accurately display the measurement results on the output BNC, the bar graph, and the numerical readout.

2.6 Outputs and Scales

2.6.1 X, Y, R, and θ Output Scales

The Sensitivity setting of the SR835 determines what input signal corresponds to 10 V full-scale output for X, Y, and R. For example, a Sensitivity of 100 mV means that a signal at f_{ref} of 100 mVrms will result in a 10 V output of R. X and Y will reach a maximum of 10 V, and a minimum of −10 V, depending upon the phase of the signal. The Sensitivity also sets the scale for the displayed bar graphs and numeric readouts.

If the signal input exceeds the sensitivity, the outputs will overload. The actual measurement is typically unaffected since it is done in floating point and has no overload. The data displayed in the chart will still be accurate.

Lock-in amplifiers are designed to measure the rms value of the ac input signal. All values of X, Y, and R outputs and displays are rms values.

θ ranges from −180° to +180° regardless of the sensitivity. When the analog output is proportional to θ, the output scale is 18°/Volt or 180° = 10 V.

2.6.2 X, Y, and R Output Offset, Ratio, and Expand

The SR835 has the ability to offset the X, Y, and R outputs. This is useful when measuring deviations in the signal around some nominal value. The offset can be set so that the output is offset to zero. Changes in the output can then be read directly from the display or output voltages. The offset is specified as a percentage of the Sensitivity and the percentage does not change when the Sensitivity is changed.

The X, Y, and R outputs may also be rescaled with a ratio function. This allows signals to be normalised to some experimental parameter being monitored with an auxiliary input voltage.

The X, Y, and R outputs may also be expanded. This takes the output (minus its offset) and multiplies by an expansion factor. Thus, a signal which is only 10% of sensitivity can be expanded to provide 10 V of output rather than only 1 V. The normal use for expand is to increase the measurement resolution around some value which is not zero. For example, suppose a signal has a nominal value of

0.9 mV and we want to measure small deviations, say 10 μV or so, in the signal. The Sensitivity of the lock-in is set to 1 mV to accommodate the nominal signal. If the offset is set to 90% of full scale, then the nominal 0.9 mV signal will result in a zero output. The 10 μV deviations in the signal only provide 100 mV of dc output. If the output is expanded by 10, these small deviations are magnified by 10 and provide outputs of 1 VDC.

The SR835 can expand the output by 10 or 100 provided the expanded output does not exceed 10 V. In the above example, the 10 μV deviations can be expanded by 100 times before they exceed full scale (at 1 mV Sensitivity).

The analog output with offset, ratio, and expand is

$$\text{Output} = \left(\frac{\frac{\text{Signal}}{\text{Sensitivity}} - \text{Offset}}{\text{Ratio In}} \right) \times \text{Expand} \times 10\ \text{V}$$

where Offset is a fraction of 1 (50% = 0.5), Expand is 1, 10, or 100, and the output cannot exceed 10 V. When ratio is disabled, Ratio In is simply 1.

In the above example (without ratio),

$$\text{Output} = (0.91\ \text{mV}\ /1\ \text{mV} - 0.9) \times 10 \times 10\ \text{V} = 1\ \text{V}$$

for a signal which is 10 μV greater than the 0.9 mV nominal (Offset = 0.9, Expand = 10).

The X and Y offset, ratio, and expand functions are output functions and do not affect the calculation of R or θ. R has its own output offset, ratio, and expand. To offset θ, simply use Auto Phase.

2.6.3 Display Behaviour

Output offsets and ratios are reflected in the displays. For example, when the X output is offset to zero, the displayed value will drop to zero also. Any display showing a quantity affected by a non-zero offset will show a highlighted Offset indicator.

Output expands do not increase the displayed numeric values of X, Y, or R. Expand increases the *resolution* of the displayed value. This is because the expand function increases the resolution of the output, not the size of the input signal. The displayed value will show increased resolution but will continue to display the original

value minus the offset. Any display showing a quantity affected by a non-unity expand will show a highlighted Expand indicator.

Output expands affect the bar graphs (they are a visualisation of the BNC outputs), but do not affect the strip charts (the charted values are floating-point numbers with full resolution available). The strip charts do reflect offsets, however.

2.7 Dynamic Reserve

Suppose the lock-in input consists of a signal at f_{ref} plus noise at some other frequencies. The real-world definition of dynamic reserve is the ratio of the largest noise signal to the actual signal (at f_{ref}). This ratio is usually expressed in dB. For example, if the f_{ref} signal is 1 μV and the noise reaches 1 mV, then the dynamic reserve is 60 dB (noise is 1000 times the signal).

Dynamic reserve is therefore a property of the input signal generated by the experiment; that is, the ratio of noise to signal at the BNC is determined by factors outside the lock-in. It is the job of the lock-in to measure the signal to the best of its ability, whatever the ratio of noise to signal at the input. Of course, it is always better to have less noise and more signal!

Unlike analog lock-in amplifiers, the SR835 with its digital phase-sensitive detectors does not suffer from dc output errors caused by large noise signals. In an analog lock-in, the PSD's nonlinearity, dc output offsets, and the high dc gain required at large reserves cause errors and drift that worsen as the reserve increases. The SR835's digital PSD has no dc output offset and its digital dc amplifier has no input offset—amplification is simply taking input numbers and multiplying by the gain. This allows the SR835 to operate at very high dynamic reserve without measurement error, output offset, or zero drift.

In the SR835, the Input Range is the only analog gain control. Set the Input Range to accommodate the largest signal present at the input, whether it is at f_{ref} or just noise. Once the Input Range is set, use the time-constant filters to achieve the best results. Then set the Sensitivity to optimise the bar graphs and numeric displays as well as the analog outputs.

2.8 The Input Amplifier

A lock-in can measure signals as small as a few nanovolts. A low-noise signal amplifier is required to boost the signal to a level where the A/D converter can digitise the signal without degrading the signal-to-noise ratio.

2.8.1 What is the Noise Floor?

The input noise of the SR835 signal amplifier determines the fundamental measurement limit. Let's set up an experiment. If the SR835's Input Range is set to its most sensitive value, then the input gain is sufficient that the noise floor of the measurement is determined by the input noise. Suppose the PSD output is low-pass filtered with a single RC filter (6 dB/oct roll-off) with a time constant of 100 ms. What will be the noise floor of the measurement?

Amplifier input noise and Johnson noise of resistors are Gaussian in nature. That is, the amount of noise is proportional to the square root of the bandwidth in which the noise is measured. A single-stage RC filter has an equivalent noise bandwidth (ENBW) of $1/(4T)$ where T is the time constant $(R \times C)$. This means that Gaussian noise is filtered with an effective bandwidth equal to the ENBW.

For Gaussian noise, the peak-to-peak noise is about 5 times the rms noise. If the noise floor is too large for your experiment, you need to add more filter stages or increase the time constant to decrease the ENBW.

All of this assumes that the signal input is being driven from a low-impedance source. Remember resistors have Johnson noise equal to $0.13\sqrt{R}$ nVrms/$\sqrt{\text{Hz}}$. Even a 50 Ω resistor has almost 1 nVrms/$\sqrt{\text{Hz}}$ of noise. A signal source with impedance in the hundreds of ohms may have Johnson noise comparable to or greater than the lock-in's input noise. To determine the overall noise of multiple noise sources, take the square root of the sum of the squares of the individual noise figures.

At lower gains (higher Input Range settings), there is not enough gain to amplify the input noise to a level greater than the noise of the A/D converter. In these cases, the apparent input noise increases with the Input Range.

2.8.2 Anti-Aliasing Filter

Between the input amplifier and A/D converter there is an anti-aliasing filter. This filter is required by the signal digitisation process. According to the Nyquist criterion, signals must be sampled at a frequency at least twice the highest signal frequency. Signals above the Nyquist frequency that reach the A/D converter would be undersampled. The result of this undersampling is "aliasing": these higher-frequency signals appear as lower frequencies in the digital data stream and would be detectable by the digital PSD.

To avoid aliasing, the analog signal is filtered to remove any signals above the lock-in's operating range. This filter has a flat passband across the operating range so as not to affect measurements. Amplitude variations and phase shifts due to this filter are calibrated at the factory and do not affect measurements. This filter is transparent to the user.

2.8.3 Input Impedance

The input impedance of the SR835 is 10 MΩ.

2.9 Input Connections

In order to achieve the best accuracy for a given measurement, care must be taken to minimise the various noise sources that can be found in the laboratory. With intrinsic noise (Johnson noise, 1/f noise, or input noise), the experiment or detector must be designed with these noise sources in mind. These noise sources are present regardless of the input connections. The effect of noise sources in the laboratory (such as motors, signal generators, etc.) and the problem of differential grounds between the detector and the lock-in can be minimised by careful input connections.

There are two basic methods for connecting a voltage signal to the lock-in. The single-ended connection is more convenient, while the differential connection eliminates spurious pickup more effectively.

2.9.1 Single-Ended Voltage Connection (A)

In the first method, the lock-in uses the A input in a single-ended mode. The lock-in detects the signal as the voltage between the centre and outer conductors of the A input only. The lock-in does not force the shield of the cable to ground; rather it is internally connected to the lock-in's ground via a resistor. The value of this resistor is selected by the user. Float uses 10 kΩ and Ground uses

10 Ω. This avoids ground-loop problems between the experiment and the lock-in due to differing ground potentials. The lock-in lets the shield "quasi-float" in order to sense the experiment ground. However, noise pickup on the shield will appear as noise to the lock-in. This is bad since the input amplifier cannot reject this noise. Common-mode noise, which appears on both the centre and shield, is rejected by the common-mode rejection of the lock-in input, but noise on only the shield is not rejected at all.

2.9.2 Differential Voltage Connection (A−B)

The second method of connection is the differential mode. The lock-in measures the voltage difference between the centre conductors of the A and B inputs. Both signal connections are shielded from spurious pickup. Noise pickup on the shields does not translate into signal noise since the shields are ignored. The shields of A and B are connected together and can be set to Float or Ground as above.

When using two cables, it is important that both cables travel the same path between the experiment and the lock-in. In particular, there should not be a large loop area enclosed by the two cables. Large loop areas are susceptible to magnetic pickup.

2.9.3 Common-Mode Signals

Common-mode signals are those signals which appear equally on both centre and shield (A mode) or on the centre of both A and B (A−B mode). With either connection scheme, it is important to minimise both the common-mode noise and the common-mode signal. Notice that the signal source is held near ground potential in both configurations. If the signal source floats at a nonzero potential, the signal which appears on both the A and B inputs will not be perfectly cancelled. The common-mode rejection ratio (CMRR) specifies the degree of cancellation. For low frequencies, a CMRR of 100 dB indicates that the common-mode signal is cancelled to 1 part in 10^5. Even with a CMRR of 100 dB, a 100 mV common-mode signal behaves like a 1 μV differential signal! This is especially bad if the common-mode signal is at the reference frequency (this often happens due to ground loops). The CMRR decreases by about 6 dB/octave (20 dB/decade) starting at around 1 kHz.

2.9.4 AC vs DC Coupling

The signal input can be either ac or dc coupled. The ac coupling passes signals above 160 mHz (0.16 Hz) and attenuates signals at lower frequencies. ac coupling also degrades the common-mode rejection of differential inputs.

At low signal frequencies, dc coupling is required.

A dc signal, if not removed by the ac coupling filter, will multiply with the reference sine wave and produce a PSD output at the reference frequency (sometimes referred to as 1f). This signal is not normally present and needs to be removed by the low-pass filter. If the dc component of the signal is large, then this output will be large and require a long time constant or synchronous filter to remove. ac coupling removes the dc component of the signal without any sacrifice in signal as long as the frequency is above 160 mHz.

2.9.5 Current Input

The current input on the SR835 uses a separate input BNC. The current input has a gain of either 10^6 or 10^8 Volts/Amp (1 μA or 10 nA range). Currents down to the femtoampere level can be measured.

The impedance of the signal source is the most important factor to consider in deciding between voltage and current measurements.

For high source impedances, greater than 1 MΩ (1 μA range) or 100 MΩ (10 nA range), and small currents, use the current input. Its relatively low input impedance greatly reduces the amplitude and phase errors caused by the cable capacitance–source impedance time constant. The cable capacitance should still be kept small to minimise the high-frequency noise gain of the current preamplifier.

For moderate to low source impedances, or larger currents, the voltage input is preferred. A small-value resistor may be used to shunt the signal current and generate a voltage signal. The lock-in then measures the voltage across the shunt resistor. Select the resistor value to keep the shunt voltage small (so it does not affect the source current) while providing enough signal for the lock-in to measure.

The current-to-voltage preamplifier is always dc coupled.

2.10 Intrinsic (Random) Noise Sources

Random noise finds its way into experiments in a variety of ways. Good experimental design can reduce these noise sources and improve the measurement stability and accuracy.

There are a variety of intrinsic noise sources which are present in all electronic signals. These sources are physical in origin.

2.10.1 Johnson Noise

Every resistor generates a noise voltage across its terminals due to thermal fluctuations in the electron density within the resistor itself. These fluctuations give rise to an open-circuit noise voltage,

$$V_{\text{noise}}(\text{rms}) = \sqrt{4kTR\Delta f}$$

where k is Boltzmann's constant $(1.38 \times 10^{-23}\ \text{J/K})$, T is the temperature in kelvins (typically 300 K), R is the resistance in ohms, and Δf is the bandwidth in hertz. Δf is the equivalent noise bandwidth of the measurement.

The amount of noise measured by the lock-in is determined by the measurement bandwidth. Remember, the lock-in does not narrow its detection bandwidth until after the phase-sensitive detectors. In a lock-in, the equivalent noise bandwidth (ENBW) of the low-pass filter (time constant) sets the detection bandwidth. In this case, the measured noise of a resistor at the lock-in input, typically the source impedance of the signal, is simply

$$V_{\text{noise}}(\text{rms}) = 0.13\sqrt{R \cdot \text{ENBW}}\ \text{nV}$$

where R is in ohms and ENBW is in hertz. The ENBW is determined by the time constant (T) and slope as shown below (for normal RC-type filters).

Slope	ENBW	Wait Time
6 dB/oct	$1/(4T)$	$5T$
12 dB/oct	$1/(8T)$	$7T$
18 dB/oct	$3/(32T)$	$9T$
24 dB/oct	$5/(64T)$	$10T$

The signal amplifier bandwidth determines the amount of broadband noise that will be amplified. This affects the Input Range (analog signal gain). The time constant sets the amount of post-

demodulation noise which will be measured at the reference frequency.

2.10.2 1/f Noise

Every 10 Ω resistor, no matter what it is made of, has the same Johnson noise. However, there is excess noise in addition to Johnson noise which arises from fluctuations in resistance due to the current flowing through the resistor. For carbon composition resistors, this is typically 0.1 μV–3 μV of rms noise per volt applied across the resistor. Metal film and wire-wound resistors have about 10 times less noise. This noise has a 1/f spectrum and makes measurements at low frequencies more difficult.

Other sources of 1/f noise include noise found in vacuum tubes and semiconductors.

2.10.3 Shot Noise

Electric current has noise due to the finite nature of the charge carriers. There is always some non-uniformity in the electron flow which generates noise in the current. This noise is called shot noise. This can appear as voltage noise when current is passed through a resistor, or as noise in a current measurement. The shot noise or current noise is given by

$$I_{\text{noise}}(\text{rms}) = \sqrt{2qI\Delta f}$$

where q is the electron charge $(1.6 \times 10^{-19}$ coulomb$)$, I is the rms ac current or dc current depending upon the circuit, and Δf is the bandwidth.

When the current input of a lock-in is used to measure an ac signal current, the bandwidth is typically so small that shot noise is not important.

2.10.4 Total Noise

All of these noise sources are incoherent. The total random noise is the square root of the sum of the squares of all the incoherent noise sources.

2.11 External Noise Sources

In addition to the intrinsic noise sources discussed previously, there are a variety of external noise sources within the laboratory.

Most of these noise sources are asynchronous, i.e. they are not related to the reference and do not occur at the reference frequency or its harmonics. Examples include lighting fixtures, motors, cooling units, radios, computer screens, etc. These noise sources affect the measurement by increasing the required dynamic reserve or lengthening the time constant.

Some noise sources, however, are related to the reference and, if picked up in the signal, will add or subtract from the actual signal and cause errors in the measurement. Typical sources of synchronous noise are ground loops between the experiment, detector, and lock-in, and electronic pickup from the reference oscillator or experimental apparatus.

Many of these noise sources can be minimised with good laboratory practice and experiment design. There are several ways in which noise sources are coupled into the signal path.

2.11.1 Capacitive Coupling

An ac voltage from a nearby piece of apparatus can couple to a detector via a stray capacitance. Although C_{stray} may be very small, the coupled noise may still be larger than a weak experimental signal. This is especially troublesome if the coupled noise is synchronous (at the reference frequency).

We can estimate the noise current caused by a stray capacitance by:

$$i = C_{\text{stray}} \frac{dV}{dt} = \omega C_{\text{stray}} V_{\text{noise}}$$

where ω is 2π times the noise frequency, V_{noise} is the noise amplitude, and C_{stray} is the stray capacitance.

For example, if the noise source is a power circuit, then $f = 60$ Hz and $V_{\text{noise}} = 120$ V. C_{stray} can be estimated using a parallel-plate equivalent capacitor. If the capacitance is roughly an area of 1 cm² separated by 10 cm, then C_{stray} is 0.009 pF. The resulting noise current will be 400 pA (at 60 Hz). This small noise current can be thousands of times larger than the signal current. If the noise source is at a higher frequency, the coupled noise will be even greater.

If the noise source is at the reference frequency, then the problem is much worse. The lock-in rejects noise at other frequencies, but pickup at the reference frequency appears as signal!

Cures for capacitive noise coupling include:

1. Removing or turning off the noise source.
2. Keeping the noise source far from the experiment (reducing C_{stray}). Do not bring the signal cables close to the noise source.
3. Designing the experiment to measure voltages with low impedance (noise current generates very little voltage).
4. Installing capacitive shielding by placing both the experiment and detector in a grounded metal box.

2.11.2 Inductive Coupling

An ac current in a nearby piece of apparatus can couple to the experiment via a magnetic field. A changing current in a nearby circuit gives rise to a changing magnetic field which induces an emf $(d\Phi_B/dt)$ in the loop connecting the detector to the experiment. This is like a transformer with the experiment–detector loop as the secondary winding.

Cures for inductively coupled noise include:

1. Removing or turning off the interfering noise source.
2. Reducing the area of the pickup loop by using twisted pairs or coaxial cables, or even twisting the two coaxial cables used in differential connections.
3. Using magnetic shielding to prevent the magnetic field from crossing the area of the experiment.
4. Measuring currents, not voltages, from high-impedance detectors.

2.11.3 Resistive Coupling or Ground Loops

Currents flowing through the ground connections can give rise to noise voltages. This is especially a problem with reference-frequency ground currents. The detector and the experiment may be grounded at different places which are at different potentials. The experiment then senses the detector signal plus the voltage due to the noise source's ground return current passing through the finite resistance of the ground between the experiment and the detector.

Cures for ground-loop problems include:

1. Grounding everything to the same physical point.
2. Using a heavy ground bus to reduce the resistance of ground connections.

3. Removing sources of large ground currents from the ground bus used for small signals.
4. Using differential excitation and signal sensing.

2.11.4 Microphonics

Not all sources of noise are electrical in origin. Mechanical noise can be translated into electrical noise by microphonic effects. Physical changes in the experiment or cables (due to vibrations for example) can result in electrical noise over the entire frequency range of the lock-in.

For example, consider a coaxial cable connecting a detector to a lock-in. The capacitance of the cable is a function of its geometry. Mechanical vibrations in the cable translate into a capacitance that varies in time, typically at the vibration frequency. Since the cable is governed by $Q = CV$, taking the derivative, we have

$$C\frac{dV}{dt} + V\frac{dC}{dt} = \frac{dQ}{dt} = i$$

Mechanical vibrations in the cable which cause a dC/dt will give rise to a current in the cable. This current affects the detector and the measured signal.

Some ways to minimise microphonic signals are:

1. Eliminate mechanical vibrations near the experiment.
2. Tie down cables carrying sensitive signals so they do not move.
3. Use a low-noise cable that is designed to reduce microphonic effects.

2.11.5 Thermocouple Effects

The emf created by junctions between dissimilar metals can give rise to many microvolts of slowly varying potentials. This source of noise is typically at very low frequency since the temperature of the detector and experiment generally changes slowly. This effect is large on the scale of many detector outputs and can be a problem for low-frequency measurements, especially in the mHz range.

Some ways to minimise thermocouple effects are:

1. Hold the temperature of the experiment or detector constant.
2. Use a compensation junction, i.e. a second junction in reverse polarity which generates an emf to cancel the thermal potential

of the first junction. This second junction should be held at the same temperature as the first junction.

2.12 Noise Measurements

Lock-in amplifiers can be used to measure noise. Noise measurements are generally used to characterise components and detectors.

The SR835 measures input signal noise at the reference frequency. Many noise sources have a frequency dependence which the lock-in can measure.

2.12.1 How is Noise Measured?

Remember that the lock-in detects signals close to the reference frequency. How close? Input signals within the detection bandwidth set by the low-pass filter time constant and roll-off appear at the output at a frequency $f = f_{\mathrm{sig}} - f_{\mathrm{ref}}$. Input noise near f_{ref} appears as noise at the output with a bandwidth of dc to the detection bandwidth.

For Gaussian noise, the equivalent noise bandwidth (ENBW) of a low-pass filter is the bandwidth of the perfect rectangular filter which passes the same amount of noise as the real filter.

The noise is simply the standard deviation (root of the mean of the squared deviations) of the measured X or Y values averaged over a period of time.

The SR835 calculates noise using floating-point math, and it is independent of the Sensitivity setting.

2.12.2 Noise Display

To display a noise measurement, choose X or Y noise (Xn or Yn) as one of the displayed parameters. The SR835 is calculating the noise all of the time, whether or not X or Y noise are being displayed. Thus, as soon as noise is displayed, the value shown is up to date and no settling time is required. If the time constant is changed, then the noise measurement will need to settle to the new value.

X and Y noise are displayed in units of V(rms). The ENBW of the time constant is not factored into the calculation. To convert to spectral noise density, divide the reading by $\sqrt{\mathrm{ENBW}}$.

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3.1 Introduction

3.1.1 Power

The power switch is on the rear panel. The SR835 is turned on by pushing the switch up.

3.1.2 Keys

The keys are grouped and labelled according to function. This manual refers to a key with square brackets such as `[Key]`. A complete description of each key follows in this chapter.

3.1.3 Touchscreen

Touchscreen buttons on the display allow for direct keypad entry, graph scaling, and more. This manual refers to touchscreen buttons with curly brackets such as `{Button}`. Touch a parameter tile to select it; the tile highlights and the `[↑]/[↓]` buttons then step through its values. Touch and hold a tile to open a numeric keypad for direct entry.

3.1.4 Reset

To reset the unit, hold down `[Reset]` while the power is turned on. The unit will use the standard settings. A similar reset is available without cycling power by pressing `[Save/Recall]` and selecting `{Recall default}`.

3.1.5 Serial Number and Firmware Version

Press and hold `[System]` to display the unit serial number and firmware version numbers. Press `[Screen Layout]` to dismiss the informational screen.

3.1.6 Local Lockout

If a computer interface has placed the unit in the Local Lockout state, indicated by the Remote LED, then the keys and touchscreen are disabled. Attempts to change the settings from the front panel will display a message indicating local control is locked out by the interface.

3.1.7 Display Off Operation

Press the `[Blank]` key to operate with the front panel display and LEDs off. The SR835 is still operating, the outputs are active, data collection continues, and the unit responds to interface commands.

To change a setting, press [Blank] to return to normal operation, change the desired parameter, then press [Blank] again.

3.2 Standard Settings

If [Reset] is held down when the power is turned on, the lock-in settings will be set to the defaults shown below rather than the settings that were in effect when the power was last turned off. The default settings may also be recalled using the *RST command over the computer interface, or by pressing [Save/Recall] and selecting {Recall default}. In this case, the communications parameters and status registers are not changed.

Parameter	Default
Reference	
Phase	0.000°
Reference Source	Internal
Internal Frequency	1.000 kHz
Harmonic #	1
Sine Amplitude	0.00 Vrms
Ext Reference Trigger	Sine
Signal (per channel)	
Input	Voltage
A–B	A
Couple	AC
Ground	Float
Current Range	1 μA
Input Range	1 V
Time Constant	100 ms
Filter Slope	6 dB/oct
Sensitivity	1 V
Outputs (per channel)	
CH Output	X (Ch 1), Y (Ch 2)
All Offsets	0.00%

All Expands	×1
All Ratios	None
All Aux Outputs	0.000 V
Display	
Green Channel	X, Graph On
Blue Channel	Y, Graph On
Yellow Channel	R, Graph On
Orange Channel	θ, Graph On
Screen Layout	Trend
Blank	No

3.3 Signal Input

3.3.1 Input `[Select]`

The Input `[Select]` key selects between the voltage and current preamplifiers for each channel. Each channel has its own independent signal input.

The voltage inputs (A and B) have a 10 MΩ, 25 pF input impedance. Their connector shields are isolated from the chassis by either 10 Ω (Ground) or 10 kΩ (Float). Do not apply more than 10 V to either input. The shields should never exceed 1 V.

The current input uses a separate I connector for each channel. The input burden resistance is 100 Ω (1 μA range) or 1 kΩ (10 nA range) to a virtual ground. The largest allowable current before overload is approximately ±4 μA or ±40 nA. No current larger than 10 mA should ever be applied to this input.

3.3.2 Voltage `[A−B]`

The voltage input can be either a single-ended (A) or differential (A−B) voltage. The shields of A and B are connected and grounded by either 10 Ω (Ground) or 10 kΩ (Float).

3.3.3 Voltage `[Couple]`

This key selects the voltage input coupling. The signal input can be either ac or dc coupled. The ac coupling option uses a high-pass filter to pass signals above 160 mHz and attenuate signals at lower frequencies. ac coupling should be used at frequencies above

160 mHz whenever possible. At lower frequencies, dc coupling is required. ac coupling results in gain and phase errors at low frequencies as well as reduced CMRR performance.

Remember, the Reference Input is ac coupled when a sine reference is used. This also results in phase errors at low frequencies.

3.3.4 Voltage `[Ground]`

This key chooses the shield grounding configuration. The shields of the input connectors (A and B) are not connected directly to the lock-in chassis ground. In Float mode, the shields are connected by 10 kΩ to the chassis ground. In Ground mode, the shields are connected by 10 Ω to ground. Typically, the shields should be grounded if the signal source is floating and floating if the signal source is grounded. Do not exceed 1 V on the shields.

3.3.5 Voltage `[Input Range]`

The voltage input range specifies the largest input signal (ac or dc coupled) before the voltage input is overloaded. The smaller the input range value, the higher the amplifier gain leading to the A/D converter. The signal-strength LEDs indicate how much of the A/D range is being used. When the lowest yellow LED is on, try decreasing the input range (increasing the gain).

In general, use the smallest input range possible without overload. Remember, the largest signal, whether it is at f_{ref} or just noise, will be the first to overload.

This setting has no effect when the current input is selected.

3.3.6 Current `[Range]`

The current input uses the I connector. The input burden resistance is 1 kΩ (10 nA range) or 100 Ω (1 μA range) to a virtual ground. The shield is chassis ground.

The current range determines the input current noise as well as the input bandwidth. The 10 nA range has 10 times lower noise but 200 times lower bandwidth. Be sure the signal frequency is below the input bandwidth limit.

Range	Input Noise	Bandwidth
1 μA	130 fA/$\sqrt{\mathrm{Hz}}$	400 kHz
10 nA	13 fA/$\sqrt{\mathrm{Hz}}$	2 kHz

The impedance of the current source should be greater than 1 MΩ when using the 1 μA range and greater than 100 MΩ when using the 10 nA range.

The current range setting has no effect when the voltage input is selected.

3.3.7 Input Overload

The Overload LED indicates an input overload. This occurs for voltage inputs greater than the voltage input range (unless removed by ac coupling) or current inputs exceeding ±4 μA (1 μA range) or ±40 nA (10 nA range).

An input overload compromises the measurement of X, Y, R, and θ. Any display of these quantities will be accompanied by an Ovld indication.

Either increase the input range or reduce the input signals to remove the overload.

Overloads can also be caused by high slew-rate voltage signals. These are large signals with fast rise or fall times (square waves with fast transitions, or large-amplitude high-frequency sine waves). In these cases, the amplifier may become slew-rate limited. When this occurs, the Overload LED may light even though the signal-strength indicator is not at its maximum. Increase the input range or reduce the input signal.

3.4 Time Constant and Filters

3.4.1 `[Time Constant]`

The time constant may be set from 1 μs to 30 ks. The time constant is indicated by 1 or 3 times 1, 10, or 100 with the appropriate units.

This time constant sets the bandwidth of the low-pass filter after the phase-sensitive detectors for X and Y. This is the filter that removes signals at frequencies other than f_{ref}. In general, longer time constants provide more noise filtering and quieter measurements but longer response times.

The time constant also determines the equivalent noise bandwidth (ENBW) of the low-pass filter. This is the measurement bandwidth for X and Y noise and depends upon the time constant and filter slope. (See the Noise discussion in the Basics chapter.)

3.4.2 Filter [Slope]

This key selects the number of poles in the low-pass filter. Choosing 6, 12, 18, or 24 dB/oct selects 1, 2, 3, or 4 poles. Using more poles can decrease the required time constant and make a measurement faster.

3.4.3 Synchronous Filter

The SR835 includes an always-active synchronous comb filter that operates across the full 1 mHz–1 MHz detection range. Unlike earlier lock-in amplifiers that required the user to enable this filter and only offered it below a few hundred hertz, the SR835 applies synchronous filtering continuously with no user action required.

The synchronous filter computes moving averages of the X and Y outputs over a complete reference period, $1/f_{\mathrm{ref}}$. In this way, outputs with periods $1/(n \times f_{\mathrm{ref}})$ average to zero and are removed. Thus frequency components in the output at $n \times f_{\mathrm{ref}}$ are all removed.

The synchronous filter follows the normal low-pass filters (time constant) and is applied at the outputs. The combination of the synchronous filter with the cascaded RC low-pass filters removes all multiples of the reference frequency and provides overall noise attenuation as well.

3.5 [Sensitivity]

The Sensitivity setting of the SR835 determines what input signal corresponds to 10 V full-scale output for X, Y, and R. For example, a Sensitivity of 100 mV means that a signal at f_{ref} of 100 mVrms will result in a 10 V output of R. X and Y will reach a maximum of 10 V depending upon the phase of the signal.

The Sensitivity also sets the scale for the displayed bar graphs and numeric readouts. Note that this is a numerical output conversion. Output overloads do not generally affect the actual measurement results! They only indicate that the output value exceeds 100% of the chosen Sensitivity and the output BNC, the bar graph, and the displayed numerical readout will be pinned at their maximums. The results displayed on the strip charts or available over the computer interfaces are the floating-point outputs and are unaffected by output overloads. Output overloads are indicated by a “Scale” indicator in the X, Y, or R displays.

The Sensitivity should be chosen to conveniently and accurately display the measurement results on the output BNC, the bar graph, and the numerical readout.

θ ranges from $-180°$ to $+180°$ regardless of the scale or range. When the analog output is proportional to θ, the output scale is 18°/Volt or 180° = 10 V.

3.6 CH1 and CH2 Outputs: Offset, Ratio, and Expand

3.6.1 Analog Outputs

Analog voltages proportional to X, Y, R, and θ can be output from the SR835. The CH1 output voltage is proportional to either X or R. The CH2 output voltage is proportional to either Y or θ. The output voltages are determined by

$$\text{Output} = \left(\frac{\frac{\text{Signal}}{\text{Sensitivity}} - \text{Offset}}{\text{Ratio In}} \right) \times \text{Expand} \times 10\ \text{V}$$

The output is normally 10 V for an input signal equal to the sensitivity. The offset subtracts a percentage of full scale (up to $\pm$999%) from the output. Expand multiplies the difference by a factor of 1, 10, or 100. This result may be divided by a ratio input (Aux Input voltage). An Aux Input voltage of 1.000 V corresponds to unity.

Outputs which would exceed $\pm$10 V generate an Output Overload and the red Overload LED will light. The actual measurement is unaffected since it is done in floating point and has no overload. Data displayed in the strip chart will still be accurate.

The X and Y offset, expand, and ratio functions are output functions —they do not affect the calculation of R or θ. R has its own output offset, expand, and ratio. Phase has no offset, ratio, or expand capability. To offset phase, use Auto Phase. To expand phase, expand the value of Y in quadrature.

3.6.2 Output `[Select]`

This key selects the source for the CH1 (CH2) output BNC. CH1 can select either X or R. CH2 can select either Y or θ. This also determines which parameter's offset and expand are adjusted with the `[Expand]` key and the Offset controls.

3.6.3 Offset

Touch `{Offset}` for the desired channel to display the offset keypad.

Touch `{Auto}` Offset to automatically adjust the X offset (or Y or R) such that the displayed value becomes zero. Offsets can also be set using the keypad or the `[↑]/[↓]` buttons.

Offsets are set as a percentage of the Sensitivity up to ±999% (10×). The offset percentage is not changed with the sensitivity—it is an output function.

Touch `{Ratio}` to cycle between Off (no ratio) and an Aux Input. The Ratio indicator is on when a ratio is being applied.

Output offsets are reflected in the displays. For example, when the X output is offset to zero, the displayed value will drop to zero also. Any display showing a quantity affected by a non-zero offset will display a highlighted Offset indicator.

3.6.4 `[Expand]`

Pressing this key selects the Expand for the current output (X or R for CH1, Y for CH2). Use the `[Select]` key to choose. θ cannot be expanded.

The expand can be ×1 (no expand), ×10, or ×100. The output can never exceed full scale (±10 V) when expanded. An output expanded beyond full scale will be output overloaded.

Output expands do not increase the displayed numeric values of X, Y, or R. Expand increases the resolution of the displayed value. The bar graphs are expanded to provide more visual resolution. The strip charts are not affected by expand (the charted values are floating-point with full resolution). Strip charts do reflect offsets, however.

3.6.5 Output Overload

Outputs which would exceed ±10 V generate an Output Overload and the red Overload LED will light. Any corresponding displayed numeric value/bar graph will indicate "Scale" in the display. The actual measurement is unaffected since it is done in floating point and has no overload.

3.7 Reference

Each data screen always displays a reference info bar across the top. This bar shows tiles displaying the reference phase, frequency, detected harmonic, and sine out amplitude. Each of these parameters can be adjusted using the controls in the Reference section of the front panel. Touching a tile brings up a numeric keypad for direct entry.

3.7.1 Source [Ref]

Press this key to cycle through the reference sources.

3.7.1.1 Internal

When the source is Internal, the SR835's synthesised internal oscillator is used as the reference. The Ref In BNC is ignored. The internal frequency is shown in the info bar. Use the [↑]/[↓] buttons or the numeric keypad to adjust the frequency.

In this mode, the Sine Out is at the internal frequency.

3.7.1.2 External

When the source is External, the SR835 will phase-lock to the external reference provided at the Ref In BNC. The SR835 will lock to frequencies between 1 mHz and 1 MHz. The external frequency is shown in the info bar.

In this mode, the Sine Out is at the external frequency.

3.7.2 Unlock

In External reference mode, the Unlock indicator turns on if the SR835 cannot lock to the external reference. This can be because the external reference amplitude is too low or the frequency is out of range.

3.7.3 External [Trig]

This key selects the external reference input trigger mode.

3.7.3.1 TTL

When either Pos TTL or Neg TTL is selected, the SR835 locks to the selected edge of a TTL square wave or pulse train. For reliable operation, the TTL signal should exceed +1.5 V when high and be less than +0.5 V when low. This trigger mode is dc coupled. This input mode should be used whenever possible since it is less noise-prone than the sine wave discriminator.

For very low frequencies (<1 Hz), a TTL reference **must** be used.

3.7.3.2 Sine

Sine input trigger locks the SR835 to the rising zero crossings of an analog signal at the Ref In BNC. This signal should be a clean sine wave at least 200 mVpp in amplitude. In this input mode, the Ref In is ac coupled (above 1 Hz).

Sine reference mode cannot be used at frequencies far below 1 Hz. At very low frequencies, the TTL input modes must be used.

3.7.4 Phase

The phase shift ranges from $-180°$ to $+180°$ with 0.001° resolution. Touch the `{Phase}` tile in the info bar to show the numeric entry screen.

When using an external reference, the reference phase shift is the phase between the external reference and the digital sine wave which is multiplying the signal in the PSD. Changing this phase shift only shifts internal sine waves. The effect of this phase shift can only be seen at the lock-in outputs X, Y, and θ. R is phase independent.

3.7.4.1 Auto Phase

Touching `{Auto}` in the numeric entry screen or pressing `[Auto Phase]` will adjust the reference phase shift so that the measured signal phase is 0°. This is done by subtracting the present measured value of θ from the reference phase shift. It will take several time constants for the outputs to reach their new values. Auto Phase may not result in a zero phase if the measurement is noisy or changing.

3.7.4.2 `[+90°]` and `[−90°]`

The `[+90°]` and `[−90°]` keys add or subtract 90.000° from the reference phase shift.

3.7.5 Frequency

If the reference source is Internal, the `[↑]/[↓]` buttons or keypad adjust the internal reference frequency. The frequency is shown in the info bar. Touch the `{Fint}` tile to show the numeric entry screen.

The internal frequency has 6 digits or 0.1 mHz resolution, whichever is larger. The frequency can range from 1.0 mHz to 1.0 MHz. The upper limit is decreased if the harmonic number is

greater than 1. In this case, the upper limit is 1 MHz/N where N is the harmonic number.

3.7.6 Harmonic Detection

The SR835 can detect signals at harmonics of the reference frequency. The SR835 multiplies the input signal with digital sine waves at $N \times f_{\mathrm{ref}}$. Only signals at this harmonic will be detected. Signals at the original reference frequency are not detected and are attenuated as if they were noise. Always check the harmonic detect number before making any measurements.

The harmonic number can be set from 1 to 99. Touch the `{Harm}` tile in the info bar to show the numeric entry screen.

The Sine Out is not affected by harmonic detection. Its frequency is always the fundamental reference frequency.

In dual-channel operation, each channel can detect at a different harmonic of the shared reference oscillator. This allows simultaneous measurement of multiple spectral components of the same signal.

3.7.7 Sine Out Amplitude

Touch the `{Ampl}` tile in the info bar to show the numeric entry screen.

The amplitude has 3 digits or 1 nV resolution, whichever is larger. The amplitude can range from 1 nVrms to 2.00 Vrms.

When the reference source is Internal, this is the excitation source provided by the SR835. When an external reference is used, this sine output provides a sine wave phase-locked to the external reference.

The TTL Out on the rear panel provides a square wave sync output at the reference frequency. This square wave is generated by discriminating the zero crossings of the sine output. This signal can provide a trigger or sync signal to the experiment when the internal reference source is used. This signal is also available when the reference is externally provided.

3.8 Display

3.8.1 `[Screen Layout]`

Press `[Screen Layout]` to cycle through the different screen layouts:

- Trend Graph
- Full-Screen Strip Chart
- Half-Screen Strip Chart
- Full-Screen FFT
- Half-Screen FFT
- Big Numbers

3.8.2 [Config]

The SR835 displays up to 4 data channels at a time, in green, blue, yellow, and orange. Each channel is assigned a parameter using the [Config] key. Parameters are chosen from X, Y, R, θ (detected), f_{int}, Reference Phase, Sine Amplitude, any Aux Input or Output, X noise, or Y noise.

Highlight one of the channel's Display tile, then select a parameter from below to assign it to the channel. Displayed parameters can be re-assigned at any time. Data is being stored for all parameters all of the time.

3.8.3 Cursor

The screen cursor is available for strip chart and FFT displays. Trend and Big Number screens do not have a cursor function.

3.8.3.1 Strip Chart Cursor

The strip chart cursor is only active when the strip chart is paused and the graph is stationary. Use {Pause} in the horizontal scale palette to stop the chart scrolling. Use the [↑]/[↓] buttons to position the cursor and read out the data values. These readouts correspond to the min, max, or mean of the data in the time bin at the cursor location. The time of the cursor location is displayed in the tile at the left edge of the scale bar below the graph. Touch this tile to switch between elapsed time and absolute time.

To read chart data relative to a reference point within the chart, position the cursor at the desired location and press the cursor confirm button. A stationary vertical line is placed at the reference location and the cursor readout values will be relative to the data at this location.

3.8.3.2 FFT Cursor

The cursor is always active in the FFT screens. The amplitude and frequency readout is at the upper right of the graph. Use the [↑]/[↓] buttons to position the cursor.

3.8.4 [Screen Shot]

Pressing [Screen Shot] saves a screen shot to the USB memory stick as a .BMP file. Set the Print Mode in the system menu. Screen is an exact screen shot, Print replaces the black background with white, and Monochrome is black and white. Screen shots also contain the time/date and summary of input parameters.

The Busy LED indicates that the USB stick is busy and should not be removed.

3.8.5 [Blank]

Press the [Blank] key to operate with the front panel display and LEDs off. The SR835 is still operating, the outputs are active, data collection continues, and the unit responds to interface commands. To change a setting, press [Blank] to return to normal operation, change the desired parameter, then press [Blank] again.

3.9 Functions

3.9.1 [Auto Phase]

[Auto Phase] adjusts the reference phase shift so that the measured signal phase is 0°. This is done by subtracting the measured value of θ from the reference phase shift. It will take several time constants for the outputs to reach their new values during which time θ will move towards 0°. Do not press [Auto Phase] again until the outputs have stabilised.

When the measurement is noisy or if the outputs are changing, Auto Phase may not result in a zero phase.

3.9.2 [Auto Range]

[Auto Range] will adjust the Input Range so that the input signal does not overload the input amplifier and that there is sufficient gain to drive the A/D converter. Remember that the signal input is both the signal at f_{ref} plus noise which may be much larger. The largest component of the signal will determine the resulting input range.

When the input is at very low frequency, `[Auto Range]` may not function well. In this case it is better to watch the signal-strength indicators and change the input range manually.

3.9.3 `[Auto Scale]`

`[Auto Scale]` will adjust the Sensitivity so that the displays and outputs of X, Y, and R are optimised. This results in numeric and bar graph displays with the best resolution possible.

Note that the Sensitivity will affect analog outputs proportional to X, Y, and R. Do not use `[Auto Scale]` if changes to the analog output scales cannot be tolerated in the experiment.

3.9.4 `[Save/Recall]`

Press `[Save/Recall]` to display the Save/Recall screen.

The SR835 can store 8 complete setups. In addition, the default setup can be recalled. Touch one of the 8 `{Save}` buttons to save a setup. Enter a name for this setup using the keypad. Touch `{Confirm}` to commit the current setup or `{Cancel}` to quit the operation. Locations which already contain a saved setup will show the setup name and the time and date it was saved.

Touch `{Recall}` for a location with a previously saved setup and a summary of settings which will change upon recall is shown. Touch `{Cancel}` to skip recalling this setup or touch `{Confirm}` to recall the setup and dismiss this screen.

3.10 Scan

3.10.1 `[Scan]` Setup

Press and hold `[Scan]` to display the scan setup screen.

The Scan Parameter can be f_{internal}, Sine Amplitude, Aux Out 1, Aux Out 2, Aux Out 3, or Aux Out 4.

The Scan Type can be Linear or Logarithmic.

The Begin and End Values are the start and stop values for the scanned parameter.

The Scan Duration is the total time to move from the Begin Value to the End Value. The Parameter Update Interval is the time spent at each scan step along the way. The shorter the time, the more steps in the scan and the smoother the parameter varies. The longer the

time, the fewer steps. With experiments that take time to settle after a parameter change, it is often beneficial to set the update time long enough to accommodate the settling.

The scan End Mode can be Repeat, Up/Down, or Once (pause at the end). When the End Mode is set to Repeat, the scan repeats over and over. When set to Once, the scan stops at the end value and the strip chart is automatically paused.

3.10.2 Running a Scan

Press `[Scan]` briefly to arm the scan. The Ready LED turns on, indicating that the scan is ready (at the begin value). The scanning parameter, if displayed on screen, is shown in orange indicating that it is under scan control.

Press `[Play/Pause]` briefly to start the scan. The Run LED turns on. The scanning parameter moves from its begin value to its end value. Press `[Play/Pause]` briefly to pause and resume the scan.

When the End Mode is Once, the scan stops at the end value, the strip chart is paused, and the Done LED turns on. Note that data collection continues while the chart is paused.

Press and hold `[Play/Pause]` to reset the scan back to the Begin Value.

3.10.3 Stopping a Scan

Press `[Scan]` briefly to turn off the scan. Turning the scan off reverts the scanned parameter back to its original value and its display back to white.

3.10.4 Changes to Scan Parameters

Once the scan is armed, the scan configuration is loaded into the scanning system for use. Edits to scan parameters (Scan Time, End Mode, etc.) can be entered from the scan setup screen but will not affect the current scan. To put any changes into action, `[Scan]` must be briefly pressed to disarm, and then briefly pressed again to re-arm. Alternatively, press and hold `[Play/Pause]` to return to the Ready state; this will also put newly changed scan parameters into effect.

3.11 [Aux Output]

Press [Aux Output] to display the Aux Output screen. The SR835 has 4 Aux Outputs on the rear panel. Use this screen to set their levels. Touch one of the {Out n} buttons to select an aux output, then use the keypad to enter a value.

3.12 System Menu

Press and hold [System] to display the system menu. The system menu is used to set non-measurement parameters such as computer interface configurations, file names, clock setup, and more.

3.12.1 Time and Date

Touch the {h} button in the Time section to highlight the hours setting. Use the keypad or [↑]/[↓] buttons to set the hour of day in 24-hour format. Continue to enter the minutes and seconds then touch {set} to commit the time. Repeat this process for the date.

The time and date are used to label data files and screen shots as well as saved settings.

3.12.2 Sounds

Sounds can be toggled from On to Mute. These are primarily the key clicks but also include various other notification sounds.

3.12.3 File Settings

Data and screenshots can be saved to a USB memory stick inserted into the front panel. Press the [Save] button to save all the data values currently plotted on the strip chart. Press the [Screen Shot] button to save an image of the screen. The memory stick must be formatted as FAT32.

File names are generated from the Prefix string and the Suffix number. The suffix number is automatically incremented after every file save. The extension is .BMP for screenshots and .CSV for data files.

3.12.4 Interface Settings

The SR835 supports USB and Ethernet interfaces. Configure the Ethernet addresses in the system menu.

The SR835 allows computers to connect via VXI-11 from a range of Trusted IP addresses or Disabled altogether. The SR835 also has a web server which allows instrument settings to be changed and outputs to be monitored from a web browser.

3.12.5 Update Firmware

The SR835's firmware can be updated via the USB memory stick. Follow the directions that accompany the update to prepare the memory stick. Insert the memory stick into the front panel and press `{Update}` to start the process.

3.13 Rear Panel

3.13.1 Power Entry Module

The power entry module is used to fuse the ac line voltage input, select the line voltage, and block high-frequency noise from entering or exiting the instrument. Refer to the Safety section of this manual for instructions on selecting the correct line voltage and fuse.

3.13.2 USB Port

The USB port allows a computer to control the SR835 via USB. The SR835 is a Test and Measurement Class instrument.

3.13.3 Ethernet Port

The Ethernet port allows a computer to control the SR835 via an ethernet network connection. In addition, the SR835 can serve web pages to the network. The network addresses and permissions are configured in the system menu.

3.13.4 Timebase In/Out

Apply a 10 MHz sine signal (1 Vrms) to the Timebase input to lock the SR835's timebase to an external timebase. The internal reference frequency of the SR835 is derived from this timebase. When multiple SR835 units are locked to the same 10 MHz timebase, harmonically related internal frequencies on different units will stay in phase indefinitely.

The Ext 10 MHz LED is on when the unit is locked to an external timebase. The SR835 also outputs its own 10 MHz timebase to another unit.

3.13.5 Aux In 1–4

These are auxiliary analog inputs which can be digitised by the SR835. The range is $\pm$10.5 V, the resolution is 1 mV, and the inputs are 1 MΩ. These inputs can be displayed on the screen, read over the computer interfaces, and graphed on the strip charts.

3.13.6 Aux Out 1–4

These are auxiliary analog outputs. The range is ± 10.5 V and the resolution is 1 mV. These outputs may be programmed from the front panel (via `[Aux Output]`), scanned (via `[Scan]`), or controlled via the computer interfaces.

3.13.7 X and Y Outputs

The X and Y lock-in outputs are always available at these rear-panel connectors. An input signal equal to the sensitivity will generate ± 10 V at these outputs. These outputs are identical to the front-panel CH1 and CH2 outputs when set to X and Y. These outputs are affected by the sensitivity and the X and Y offsets and expands.

3.13.8 Signal Monitor Output

This BNC provides a buffered output from the signal amplifier. This is the signal just before the anti-aliasing filter, A/D converter, and PSD. The amplifier gain is determined by the Input Range (voltage input) or the Current Range (current input). The output is about 1 V for a maximum input.

This output is useful for determining the cause of input overloads, not necessarily for viewing the signal of interest. If the f_{ref} signal is small relative to the input range, it may not be amplified enough to be viewed at the monitor output.

3.13.9 Trig Input

This TTL input is used to start data capture to the internal buffer. This buffer is only accessible via the computer interface. See the Remote Programming chapter for more information.

3.13.10 TTL Out

This output is the TTL sync output for the internal oscillator. The output is a square wave whose edges are linked to the sine wave zero crossings. This is useful when the sine output amplitude is small and a synchronous trigger is required (for a scope, for example). This output is active even when locked to an external reference.

3.13.11 Preamp Connector

This 9-pin connector provides power to external preamplifiers such as the SR550 or SR551.

Pin	Voltage
1	+18.5 V (up to 100 mA)
2	+5 V (up to 100 mA)
6	−18.5 V (up to 100 mA)
7	Signal Ground
8	Ground

3.13.11.1 Using SRS Preamps

When using an SRS preamp, connect the power cable (standard male-to-male 9-pin D connectors) from the preamp to the rear panel preamp connector on the SR835. Use BNC cables to connect the A output from the preamp to the A input of the SR835 and the B output from the preamp (preamp ground) to the B input of the SR835. Use A−B as the input configuration. Be sure to twist the A and B cables so that there is no differential noise pickup between the cables.

The signal coupling and input impedance is now determined by the preamp.

The SR835 does not compensate for the gain of the preamp. Measurements made with a preamp need to be divided by the gain of the preamp.

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4.1 Introduction

The SR835 Dual Lock-In Amplifier may be remotely programmed via the USB or Ethernet interfaces. Both interfaces accept commands at all times; the SR835 sends responses only to the interface that issued the query.

4.2 Communicating with USB

The SR835 appears as a USB Test and Measurement Class (USBTMC) device. It interfaces directly with NI-VISA, LabVIEW, MATLAB, and Python (via `pyvisa`). No special driver installation is required on modern operating systems.

Connect the SR835 to a host computer using a standard USB Type-B cable to the rear-panel USB connector.

4.3 Communicating with Ethernet

The SR835 supports VXI-11 (TCP/IP Instrument Protocol) and raw TCP socket connections. It interfaces with NI-VISA, LabVIEW, MATLAB, and any TCP-capable programming environment.

Before communicating over Ethernet, configure the network settings in the System menu:

- **IP Mode** — Select DHCP (automatic) or Static.
- **Static IP** — If using static addressing, set the IP address, subnet mask, and gateway.
- The current IP address is displayed in the System menu and can be queried with `SYSTem:ETHernet:IP?`.

The VXI-11 service runs on TCP port 111 (portmapper). Raw TCP socket connections use port 5025 by default.

Data streaming uses a separate UDP or TCP connection; see the Streaming section below.

4.4 Status Indicators

Three interface status indicators assist with programming:

Activity Flashes whenever any interface is actively sending or receiving data.

Error Flashes when the SR835 detects a command error, such as an unrecognized command or an out-of-range parameter.

Remote Illuminates whenever the SR835 is in remote mode (front panel locked out).

4.5 SCPI Command Syntax

The SR835 uses SCPI (Standard Commands for Programmable Instruments) syntax. A command consists of a keyword path, optional arguments, and a terminator.

4.5.1 Command Structure

A typical command looks like:

```
SENSe:CHANnel1:TC 100MS
```

The command tree is navigated with colons separating keywords. A space separates the last keyword from the argument(s). Multiple arguments are separated by commas. The command terminator is a linefeed (LF, `0x0A`).

Multiple commands may be sent on one line by separating them with semicolons:

```
SENSe:CHANnel1:TC 100MS;SENSe:CHANnel1:SLOPe 24
```

4.5.2 Queries

The current value of any settable parameter may be read by appending a question mark to the command keyword. No space is permitted between the keyword and the question mark:

```
SENSe:CHANnel1:TC?
```

Query responses are returned as ASCII strings terminated by linefeed. If multiple queries are sent on one line, responses are separated by semicolons.

4.5.3 Case and Abbreviation

Commands and arguments are case-insensitive. Keywords may be abbreviated to the uppercase portion shown in the command reference. For example, `SENSe` may be sent as `SENS`, `sense`, or `sense`. The full keyword or its short form are both accepted; partial abbreviations other than the documented short form are not.

4.5.4 Numeric Arguments

Integer arguments must be expressed without a decimal point. Real-number arguments may be integer, floating-point, or exponential format (e.g., `5`, `5.0`, `.5E1`).

For frequency, voltage, and phase parameters, an optional unit suffix may be appended:

Frequency	Voltage	Phase
HZ	NV (nanovolts)	UDEG (microdegrees)
KHZ	UV (microvolts)	MDEG (millidegrees)
MHZ	MV (millivolts)	DEG (degrees)
	V (volts)	RAD, MRAD, URAD

If the unit suffix is omitted, the base unit is assumed (Hz, V, or degrees). For example, the following commands are equivalent:

```
SOURce:CHANnel1:FREQuency 10000
SOURce:CHANnel1:FREQuency 10 KHZ
SOURce:CHANnel1:FREQuency 10E3
```

Queries always return values in the base unit.

4.5.5 Enumerated Arguments

Many commands accept enumerated string arguments. The argument may be the short form (uppercase letters in the reference) or the full string, in any case. For example, `INPut:CHANnel1:COUPling` accepts:

`AC` or `ac`
`DC` or `dc`

4.6 Interface Ready and Status

When a command is received, the SR835 clears the Interface Ready bit (bit 1) in the Status Byte, processes the command, then sets the bit again. Most commands execute in under 1 ms, so the host rarely needs to poll this bit.

Send `*OPC?` after a slow command; the `1` response confirms completion.

4.7 Status Reporting

The SR835 implements the IEEE 488.2 status reporting model with additional instrument-specific registers.

4.7.1 Status Byte

The Status Byte summarizes the state of all status registers:

Bit	Name	Meaning

0	LIA	Lock-in status event (see LIA Status)
1	Interface Ready	Ready to accept a new command
2	Error	Error status event (see Error Status)
3	Questionable	Questionable data condition
4	MAV	Message available (response in output queue)
5	ESB	Standard Event Status summary
6	MSS	Master Status Summary
7	Operation	Operation status event

Query the Status Byte with `*STB?`. Enable Service Request generation with `*SRE <mask>`.

4.7.2 Standard Event Status Register

Bit	Name	Meaning
0	OPC	Operation Complete
1	(unused)	
2	QYE	Query Error (response with no listener)
3	DDE	Device-Dependent Error
4	EXE	Execution Error (parameter out of range)
5	CME	Command Error (syntax/parse error)
6	(unused)	
7	PON	Power-On (set once after power cycle)

Query with `*ESR?` (reads and clears). Set the enable mask with `*ESE <mask>`. When (ESR & ESE) is non-zero, bit 5 (ESB) is set in the Status Byte.

4.7.3 Lock-In Status Word

The lock-in status word reports instrument-specific conditions:

Bit	Name	Meaning
0	Ch-A Input Overload	Input amplifier overload on channel A
1	Ch-A Filter Overload	Filter/output overload on channel A
2	Ch-A Output Overload	Analog output overload on channel A
3	Ch-B Input Overload	Input amplifier overload on channel B
4	Ch-B Filter Overload	Filter/output overload on channel B
5	Ch-B Output Overload	Analog output overload on channel B
6	Ch-A Reference Unlock	Ch-A PLL is unlocked
7	Ch-B Reference Unlock	Ch-B PLL is unlocked

Query with `STATus:LIA?`. Set the enable mask with `STATus:LIA:ENAble <mask>`. When (LIA & Enable) is non-zero, bit 0 in the Status Byte is set.

4.7.4 Error Status Byte

Bit	Name	Meaning
0	Print/Plot Error	Hard-copy output error
1	Backup Error	Settings memory error
2	USB Error	USB communication error
3	Ethernet Error	Network communication error
4	(unused)	
5	RAM Error	Self-test memory error
6	ROM Error	Self-test firmware error
7	(unused)	

Query with `STATus:ERRor?`. Set the enable mask with `STATus:ERRor:ENAble <mask>`. When (Error & Enable) is non-zero, bit 2 in the Status Byte is set.

4.7.5 Clearing Status

`*CLS` clears all status event registers and the error queue. The enable registers are not affected.

`STATus:PRESet` resets all enable registers to their default (zero) values.

`*PSC 1` causes the status enable registers to be cleared on power-up, preventing stale SRQ assertions. `*PSC 0` preserves enable registers across power cycles.

4.8 Data Capture

The SR835 has an internal data capture buffer for recording time-series measurements.

4.8.1 Configuration

Set the quantities to capture with `CAPTure:CONFig`:

- `X` — X only (one channel, maximum buffer length)
- `XY` — X and Y
- `RT` — R and theta
- `XYRT` — X, Y, R, and theta (four channels, 1/4 buffer length per channel)

Set the buffer size with `CAPTure:LENgth` (2–4096 kbytes) and the sample rate with `CAPTure:RATE`.

4.8.2 Capture Modes

Set the capture mode with `CAPTure:MODE`:

- `ONEShot` — Capture stops when the buffer is full.
- `LOOP` — Capture wraps around, continuously overwriting the oldest data.
- `LOG` — Sample rate decreases as the buffer fills, providing a logarithmic time record.

4.8.3 Triggering and Control

Enable trigger-start with `CAPTure:TRIGger ON`. When armed, capture begins on the next rising edge of the channel-A reference input.

Start capture with `CAPTure:STARt`. Stop with `CAPTure:STOP`. Query the current state with `CAPTure:STATus?` (returns `IDLE`, `ARMED`, `RUNNING`, or `FULL`).

4.8.4 Reading Data

Query the number of stored points with `CAPTure:POINts?`. Read data with:

```
CAPTure:DATA? <first>,<count>
```

This returns *count* samples starting at index *first* as an IEEE 754 binary float block. The host must parse the IEEE 488.2 definite length block header.

4.8.5 Example: Capture 1000 Points

```
CAPTure:CONFig XY
CAPTure:LENgth 64
CAPTure:RATE 256
CAPTure:MODE ONES
CAPTure:STARt
*OPC?
```
...wait for 1 response...
```
CAPTure:POINts?
CAPTure:DATA? 0,1000
```

4.9 Data Streaming

The SR835 can stream measurement data continuously over Ethernet at rates up to the full demodulation bandwidth. Streaming provides real-time data to the host without the capacity limits of the capture buffer.

4.9.1 Configuration

Configure streaming with:

- `STReam:CONFig` — Set channel configuration (`X`, `XY`, `RT`, `XYRT`).
- `STReam:RATE` — Set the rate divisor exponent (0–15). The stream rate is the base demodulation rate divided by 2^n.
- `STReam:FORMat` — Choose `FLOAT32` (IEEE 754) or `INT16` (scaled integers, lower bandwidth).
- `STReam:PACKet` — Set packet size (128, 256, 512, or 1024 bytes).
- `STReam:PORT` — Set the destination port (1024–65535).
- `STReam:ENDian` — Set byte order (`LITTle` or `BIG`).

4.9.2 Starting and Stopping

Enable streaming with `STReam:ENAble ON`. The SR835 begins sending UDP packets to the host IP address that issued the command.

Stop streaming with `STReam:ENAble OFF`.

4.9.3 Data Format

Each packet contains a sequence of samples in the configured format. For `FLOAT32`, each sample is a 4-byte IEEE 754 float per configured channel. For `INT16`, each sample is a 2-byte signed integer per channel, scaled so that the current sensitivity equals $2^{\{15\}} - 1$.

The host must keep up with the data rate. If the host falls behind, packets are dropped without notification.

4.10 Programming Examples

The following examples illustrate common programming tasks using SCPI commands.

4.10.1 Basic Measurement

Configure channel A for a voltage measurement at 10 kHz with a 100 ms time constant and 24 dB/oct filter slope:

```
*RST
INPut:CHANnel1:MODE A
INPut:CHANnel1:COUPling AC
REFerence:CHANnel1:SOURce INT
SOURce:CHANnel1:FREQuency 10 KHZ
SOURce:CHANnel1:VOLTage 1
SENSe:CHANnel1:TC 100MS
SENSe:CHANnel1:SLOPe 24
```

Read the measurement:

```
READ:CHANnel1?
```

This returns X, Y, R, and theta as four comma-separated values.

4.10.2 Dual-Channel Measurement

Configure both channels for simultaneous measurement at different frequencies:

```
SOURce:CHANnel1:FREQuency 1 KHZ
SOURce:CHANnel1:VOLTage 0.5
SOURce:CHANnel2:FREQuency 10 KHZ
SOURce:CHANnel2:VOLTage 0.5
SENSe:CHANnel1:TC 300MS
SENSe:CHANnel2:TC 30MS
```

Read both channels simultaneously:

```
SNAP?
```

This returns X_A, Y_A, R_A, theta_A, X_B, Y_B, R_B, theta_B.

4.10.3 External Reference with Harmonic Detection

Lock to the 2nd harmonic of an external TTL reference:

```
REFerence:CHANnel1:SOURce EXT
REFerence:CHANnel1:TRIGger POS
REFerence:CHANnel1:HARMonic 2
```

Verify the lock and read the detection frequency:

```
REFerence:CHANnel1:UNLock?
REFerence:CHANnel1:FREQuency?
```

4.10.4 Using Service Request (SRQ)

Set up the SR835 to assert SRQ when a measurement is complete:

```
*CLS
*ESE 1
*SRE 32
```

This enables the OPC bit in the Event Status register and routes it to the Service Request line. After issuing a slow command, send `*OPC` (no query). The SR835 sets the MSS bit when the operation finishes. The host can poll `*STB?` to detect the event.

4.10.5 Aux Output Proportional to Measurement

Route the channel-A R value to Aux Output 1, with a full-scale mapping of 1 V input to ±10 V output:

```
AUXiliary:OUTPut1:MODE MEAS
AUXiliary:OUTPut1:MEASure:SOURce RA
AUXiliary:OUTPut1:MEASure:FSCale 1
AUXiliary:OUTPut1:MEASure:OFFSet 0
AUXiliary:OUTPut1:MEASure:EXPand 1
```

5 Remote Commands

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5.1 Common Commands (IEEE 488.2)

`*IDN?`

Returns the instrument identification string in the format *manufacturer, model, serial, firmware version*.

Returns:
Four comma-separated fields, e.g. `Stanford_Research_Systems,SR835,s/n012345,v1.0.0`.

Examples:

`*IDN?` *returns identification string*

`*RST`

Resets all instrument parameters to factory defaults. The interface settings (IP configuration, etc.) are not affected.

Examples:

`*RST`

`*OPC?`

Returns `1` when all pending operations have completed. Use this after a slow command (e.g.
Auto Phase) to synchronize the host.

Returns:
`1` when ready.

Examples:

`*OPC?` *wait for operation to complete*

`*CLS`

Clears all status event registers and the error queue. The enable registers are not affected.

Examples:

`*CLS`

`*TST?`

Executes the internal self-test and returns the result.

Returns:
`0` if the self-test passed, or a non-zero error code.

Examples:

`*TST?` *returns 0 if OK*

`*SAV <n>`

Saves the current instrument state to memory slot *n*.

Arguments:
`<n>` — integer, 1–9

Examples:

`*SAV 3` *save current state to slot 3*

`*RCL <n>`

Recalls a previously saved instrument state from slot *n*.

Arguments:
`<n>` — integer, 1–9

Examples:

`*RCL 3` *recall state from slot 3*

`*ESE(?) <mask>`

Sets or queries the Standard Event Status Enable register. Enabled bits allow the corresponding events in the ESR to set bit 5 (ESB) in the Status Byte.

Arguments:
`<mask>` — integer, 0–255

Returns:
The current enable mask.

Examples:

`*ESE 60` *enable bits 2--5 (QYE, DDE, EXE, CME)*
`*ESE?` *query the enable mask*

`*ESR?`

Queries and clears the Standard Event Status Register. Each bit indicates whether the corresponding event has occurred since the last `*ESR?` or `*CLS`.

Returns:
Integer, 0–255.

Examples:

`*ESR?` *read and clear ESR*

`*SRE(?) <mask>`

Sets or queries the Service Request Enable register. Enabled bits allow the corresponding Status Byte conditions to set the MSS bit in the Status Byte.

Arguments:
`<mask>` — integer, 0–255

Returns:
The current enable mask.

Examples:

`*SRE 32` *enable SRQ on ESB (bit 5)*

`*STB?`

Queries the Status Byte register without clearing it.

Returns:
Integer, 0–255.

Examples:

`*STB?` *read status byte*

`*PSC(?) <b>`

Sets or queries the Power-on Status Clear flag. When set to `1`, all status enable registers are cleared at power-up, preventing stale SRQ assertions.

Arguments:
<b> — 0 (preserve enables across power cycles) or 1 (clear enables at power-up)

Returns:
0 or 1.

Examples:

*PSC 1 *clear enables at power-up*

5.2 INPut Subsystem

In all INPut commands, <n> is the channel number: 1 (channel A) or 2 (channel B).

INPut:CHANnel<n>:MODE(?) <m>

Sets or queries the input topology for channel *n*.

Arguments:
<m> — A (single-ended voltage), AB (differential voltage), or I (current)

Returns:
Current mode string.

Examples:

INPut:CHANnel1:MODE AB *set channel A to differential*
INPut:CHANnel1:MODE? *returns AB*

INPut:CHANnel<n>:COUPling(?) <c>

Sets or queries the input coupling.

Arguments:
<c> — AC or DC

Returns:
Current coupling string.

Examples:

INPut:CHANnel1:COUPling AC *set AC coupling*

INPut:CHANnel<n>:SOURce(?) <s>

Selects or queries which physical input connectors are used.

Arguments:
<s> — FRONt (In-1/In-2, front panel) or REAR (In-3/In-4, rear panel)

Returns:
Current source string.

Examples:

INPut:CHANnel1:SOURce FRONT *use front-panel inputs*

INPut:CHANnel<n>:CURRent:GAIN(?) <g>

Sets or queries the current-input transimpedance gain. Only relevant when the input mode is I.

Arguments:
<g> — 100K, 1MEG, 10MEG, or 100MEG (ohms)

Returns:
Current gain string.

Examples:

INPut:CHANnel1:CURRent:GAIN 1MEG *set 1 MOhm gain*

INPut:IN<k>:SHIeld(?) <g>

Sets or queries the BNC shield connection for physical input *k*.

Arguments:
<k> — input number, 1–4
<g> — FLOat or GROund

Returns:
Current shield setting.

Examples:

INPut:IN1:SHIeld GRO *ground the shield on input 1*

INPut:IN<k>:GAIN(?) <g>

Sets or queries the hardware preamp gain for physical input *k*.

Arguments:
<k> — input number, 1–4
<g> — 1, 2, 4, 8, 16, 32, 64, 128, or 256

Returns:
Current gain as an integer.

Examples:

`INPut:IN1:GAIN 16` *set preamp gain to 16*

5.3 SENSe Subsystem

`SENSe:CHANnel<n>:TC(?) <t>`

Sets or queries the demodulator time constant for channel *n*.

Arguments:
`<t>` — time constant value with optional unit suffix; range 1 μs to 86400 s; accepted suffixes: `US`, `MS`, `S`, `KS`; rounded to the nearest available 1–3–10 step

Returns:
Time constant in seconds.

Examples:

`SENSe:CHANnel1:TC 100MS` *set 100 ms time constant*
`SENSe:CHANnel1:TC?` *returns 0.1*

`SENSe:CHANnel<n>:SLOPe(?) <s>`

Sets or queries the low-pass filter roll-off slope.

Arguments:
`<s>` — `6`, `12`, `18`, or `24` (dB/octave)

Returns:
Current slope in dB/oct.

Examples:

`SENSe:CHANnel1:SLOPe 24` *set 24 dB/oct roll-off*

`SENSe:CHANnel<n>:FILTer:NOTCH(?) <f>`

Sets or queries the line-frequency notch filter configuration.

Arguments:
`<f>` — `OFF` (disabled), `LINE` (notch at line frequency), `2LINE` (notch at 2×line), or `BOTH` (notches at both)

Returns:
Current notch setting.

Examples:

`SENSe:CHANnel1:FILTer:NOTCH BOTH` *enable both line notches*

`SENSe:CHANnel<n>:FILTer:BPASs(?) <b>`

Enables or queries the auto-tracking bandpass filter ahead of the PSD.

Arguments:
`<b>` — `ON` or `OFF`

Returns:
Current state.

Examples:

`SENSe:CHANnel1:FILTer:BPASs ON` *enable bandpass filter*

`SENSe:CHANnel<n>:PHASe(?) <p>`

Sets or queries the reference phase offset for channel *n*.

Arguments:
`<p>` — phase in degrees (default) or with unit suffix; range −360000 to +360000 degrees; wraps at ±180°

Returns:
Phase offset in degrees.

Examples:

`SENSe:CHANnel1:PHASe 90` *set 90 degree offset*
`SENSe:CHANnel1:PHASe 1.5708 RAD` *equivalent in radians*

`SENSe:CHANnel<n>:PHASe:UNIT(?) <u>`

Sets or queries the display unit for phase readout.

Arguments:
`<u>` — `DEG`, `MDEG`, `UDEG`, `RAD`, `MRAD`, or `URAD`

Returns:
Current unit string.

Examples:

`SENSe:CHANnel1:PHASe:UNIT RAD` *display phase in radians*

`SENSe:CHANnel<n>:PHASe:AUTO`

Executes the Auto Phase function: adjusts the phase offset so that the current signal appears entirely in the X channel ($Y \approx 0$). The outputs require several time constants to settle after this command.

Examples:

`SENSe:CHANnel1:PHASe:AUTO` *auto-phase channel A*

`SENSe:CHANnel<n>:ADCLevel?`

Queries the ADC input utilization level for channel *n*.

Returns:
Integer 1–5, where each step represents approximately 20% of the ADC range. A reading of 5 indicates the ADC is near full scale.

Examples:

`SENSe:CHANnel1:ADCLevel?` *check ADC utilization*

5.4 REFerence Subsystem

`REFerence:CHANnel<n>:SOURce(?) <s>`

Sets or queries the reference source for channel *n*.

Arguments:
`<s>` — `INTernal` (use internal oscillator), `EXTernal` (lock to external reference input), or `DUAL` (dual-reference mode: detection at $N_{\text{int}} f_{\text{int}} - N_{\text{ext}} f_{\text{ext}}$)

Returns:
Current source string.

Examples:

`REFerence:CHANnel1:SOURce EXT` *lock to external reference*

`REFerence:CHANnel<n>:TRIGger(?) <t>`

Sets or queries how the external reference input is interpreted.

Arguments:
<t> — SINE (zero-crossing), POS (TTL rising edge), or NEG (TTL falling edge); for frequencies below 1 Hz, a TTL reference must be used

Returns:
Current trigger type.

Examples:

REFerence:CHANnel1:TRIGger POS *use TTL rising edge*

REFerence:CHANnel<n>:SHIeld(?) <g>

Sets or queries the reference-input BNC shield connection.

Arguments:
<g> — FLOat or GROund

Returns:
Current setting.

Examples:

REFerence:CHANnel1:SHIeld FLO *float the reference shield*

REFerence:CHANnel<n>:HARMonic(?) <h>

Sets or queries the detection harmonic. The lock-in detects at $h \times f_{\mathrm{ref}}$, which must not exceed 1 MHz.

Arguments:
<h> — integer, 1–99

Returns:
Current harmonic number.

Examples:

REFerence:CHANnel1:HARMonic 2 *detect at 2nd harmonic*

REFerence:CHANnel<n>:DUAL:HARMonic(?) <h>

Sets or queries the external-reference harmonic used in dual-reference mode.

Arguments:
<h> — integer, 1–99

Returns:
Current dual harmonic number.

Examples:

`REFerence:CHANnel1:DUAL:HARMonic 1` *use fundamental of external ref*

`REFerence:CHANnel<n>:FREQuency?`

Queries the actual locked reference frequency for channel *n*. In internal mode, returns the internal oscillator frequency. In external mode, returns the measured external frequency.

Returns:
Frequency in Hz.

Examples:

`REFerence:CHANnel1:FREQuency?` *read locked frequency*

`REFerence:CHANnel<n>:UNLock?`

Queries whether the phase-locked loop is locked.

Returns:
`0` if locked, `1` if unlocked.

Examples:

`REFerence:CHANnel1:UNLock?` *check PLL lock status*

5.5 SOURce Subsystem

The SOURce subsystem controls the internal sine-wave oscillator and output for each channel.

`SOURce:CHANnel<n>:FREQuency(?) <f>`

Sets or queries the internal oscillator frequency.

Arguments:
`<f>` — frequency value, 1 mHz to 1 MHz, with optional unit suffix (`HZ`, `KHZ`, `MHZ`); rounded to 6 significant digits or 0.1 mHz, whichever is greater

Returns:
Frequency in Hz.

Examples:

`SOURce:CHANnel1:FREQuency 10 KHZ` *set frequency to 10 kHz*
`SOURce:CHANnel1:FREQuency?` *returns 10000.0*

`SOURce:CHANnel<n>:VOLTage(?) <v>`

Sets or queries the sine output amplitude (differential, open-circuit).

Arguments:
`<v>` — amplitude, 1 nV to 2 Vrms, with optional unit suffix (`NV`, `UV`, `MV`, `V`)

Returns:
Amplitude in Vrms.

Examples:

`SOURce:CHANnel1:VOLTage 100 MV` *set 100 mVrms output*

`SOURce:CHANnel<n>:OFFSet(?) <v>`

Sets or queries the DC offset on the sine output.

Arguments:
`<v>` — offset voltage, -5 V to $+5$ V

Returns:
Offset in volts.

Examples:

`SOURce:CHANnel1:OFFSet 0.5` *set 0.5 V DC offset*

`SOURce:CHANnel<n>:SHIeld(?) <g>`

Sets or queries the sine output BNC shield connection.

Arguments:
`<g>` — `FLOat` or `GROund`

Returns:
Current setting.

Examples:

`SOURce:CHANnel1:SHIeld GRO` *ground the output shield*

SOURce:CHANnel<n>:PRESet:FREQuency(?) <slot>,<f>

Stores or recalls a frequency preset.

Arguments:
`<slot>` — 1–6
`<f>` — frequency, 1 mHz to 1 MHz

Returns:
(query) Stored frequency in Hz.

Examples:

`SOURce:CHANnel1:PRESet:FREQuency 1,1000` *store 1 kHz in slot 1*
`SOURce:CHANnel1:PRESet:FREQuency? 1` *returns 1000.0*

SOURce:CHANnel<n>:PRESet:VOLTage(?) <slot>,<v>

Stores or recalls an amplitude preset.

Arguments:
`<slot>` — 1–6
`<v>` — amplitude, 1 nV to 2 Vrms

Returns:
(query) Stored amplitude in Vrms.

Examples:

`SOURce:CHANnel1:PRESet:VOLTage 1,0.1` *store 100 mVrms in slot 1*

SOURce:CHANnel<n>:PRESet:OFFSet(?) <slot>,<v>

Stores or recalls a DC offset preset.

Arguments:
`<slot>` — 1–6
`<v>` — −5 V to +5 V

Returns:
(query) Stored offset in volts.

Examples:

`SOURce:CHANnel1:PRESet:OFFSet 1,0.0` *store 0 V offset in slot 1*

SOURce:CHANnel<n>:PRESet:LOAD <slot>

Atomically applies the frequency, amplitude, and DC offset stored in the given preset slot. All three parameters change simultaneously to avoid transient conditions.

Arguments:
`<slot>` — 1–6

Examples:

`SOURce:CHANnel1:PRESet:LOAD 1` *apply all presets from slot 1*

5.6 OUTPut and READ Subsystem

`OUTPut:CHANnel<n>:QUANtity(?) <q>`

Sets or queries which quantity is routed to the channel *n* analog output.

Arguments:
`<q>` — `X`, `Y`, `R`, `THETa`, `XNOise`, `YNOise`, or `FREQuency`

Returns:
Current quantity string.

Examples:

`OUTPut:CHANnel1:QUANtity R` *route R to channel 1 output*

`OUTPut:CHANnel<n>:REL:ENAble(?) <b>`

Enables or disables REL (relative/baseline subtraction) mode on channel *n*.

Arguments:
`<b>` — `ON` or `OFF`

Returns:
Current state.

Examples:

`OUTPut:CHANnel1:REL:ENAble ON` *enable REL mode*

`OUTPut:CHANnel<n>:REL:CLEar`

Clears the stored REL baseline value for channel *n*, resetting it to zero.

Examples:

`OUTPut:CHANnel1:REL:CLEar` *reset baseline to zero*

`READ:CHANnel<n>?`

Returns the current X, Y, R, and theta values for channel *n* as four comma-separated values. This is a synchronous read: all four values are sampled at the same instant.

Returns:
`X,Y,R,theta` (volts, volts, volts, degrees).

Examples:

`READ:CHANnel1?` *returns e.g. 1.234e-3,5.678e-5,1.235e-3,2.63*

`READ:CHANnel<n>:X? / :Y? / :R? / :THETa?`

Returns a single measurement quantity for channel *n*.

Returns:
Value in base units (volts or degrees).

Examples:

`READ:CHANnel1:R?` *returns R in volts*

`READ:CHANnel<n>:XNOise? / :YNOise?`

Returns the equivalent input noise spectral density measured at the demodulator output.

Returns:
Noise in $V/\sqrt{Hz}$.

Examples:

`READ:CHANnel1:XNOise?` *returns noise density in V/rtHz*

`READ:CHANnel<n>:FREQuency?`

Returns the current locked reference frequency for channel *n*.

Returns:
Frequency in Hz.

Examples:

`READ:CHANnel1:FREQuency?` *read detection frequency*

`SNAP?`

Returns X, Y, R, and theta for both channels simultaneously. All eight values are sampled at the same instant, ensuring phase coherence between channels.

Returns:
`X_A,Y_A,R_A,theta_A,X_B,Y_B,R_B,theta_B`.

Examples:

`SNAP?` *simultaneous dual-channel read*

5.7 AUXiliary Subsystem

In AUXiliary commands, `<m>` is the aux output number: `1` or `2`.

`AUXiliary:OUTPut<m>:VOLTage(?) <v>`

Sets or queries the aux output voltage when in Fixed mode.

Arguments:
`<v>` — -10.5 V to $+10.5$ V

Returns:
Programmed voltage in volts.

Examples:

`AUXiliary:OUTPut1:VOLTage 5.0` *set Aux 1 to 5.0 V*

`AUXiliary:OUTPut<m>:MODE(?) <mo>`

Sets or queries the aux output operating mode.

Arguments:
`<mo>` — `FIXed` (manual voltage), `MEASurement` (proportional to a lock-in output), or `SCAN` (swept by the scan engine)

Returns:
Current mode string.

Examples:

`AUXiliary:OUTPut1:MODE MEAS` *route a measurement to the aux output*

AUXiliary:OUTPut<m>:MEASure:SOURce(?) <s>

Sets or queries which measurement drives the aux output in Measurement mode.

Arguments:
<s> — XA, YA, RA, TA, XB, YB, RB, or TB

Returns:
Current source string.

Examples:

`AUXiliary:OUTPut1:MEASure:SOURce RA` *output channel-A R value*

AUXiliary:OUTPut<m>:MEASure:FSCale(?) <v>

Sets or queries the measurement value that maps to ±10 V on the aux output.

Arguments:
<v> — positive real number in the measurement's base unit

Returns:
Full-scale value.

Examples:

`AUXiliary:OUTPut1:MEASure:FSCale 0.001` *1 mV full scale*

AUXiliary:OUTPut<m>:MEASure:OFFSet(?) <p>

Sets or queries the output offset as a percentage of full scale.

Arguments:
<p> — −999 to +999 (percent)

Returns:
Offset in percent.

Examples:

`AUXiliary:OUTPut1:MEASure:OFFSet 0` *zero offset*

AUXiliary:OUTPut<m>:MEASure:EXPand(?) <x>

Sets or queries the output expand multiplier.

Arguments:
<x> — 1, 2, 4, 8, 16, 32, 64, 128, or 256

Returns:
Current expand factor.

Examples:

`AUXiliary:OUTPut1:MEASure:EXPand 10` *10x expand*

5.8 CAPTure Subsystem

`CAPTure:CONFig(?) <c>`

Sets or queries the capture channel configuration.

Arguments:
<c> — `X` (X only), `XY` (X and Y), `RT` (R and theta), or `XYRT` (all four); more channels reduce the maximum number of points

Returns:
Current configuration string.

Examples:

`CAPTure:CONFig XY` *capture X and Y*

`CAPTure:LENgth(?) <n>`

Sets or queries the capture buffer size.

Arguments:
<n> — 2 to 4096 (kilobytes)

Returns:
Buffer size in kilobytes.

Examples:

`CAPTure:LENgth 256` *256 kB buffer*

`CAPTure:RATE(?) <r>`

Sets or queries the capture sample rate.

Arguments:
<r> — sample rate in Hz

Returns:
Actual sample rate in Hz.

Examples:

`CAPTure:RATE 1024` *capture at 1024 Sa/s*

`CAPTure:MODE(?) <m>`

Sets or queries the capture mode.

Arguments:
`<m>` — `ONEShot` (stop when full), `LOOP` (circular buffer), or `LOG` (logarithmic time spacing)

Returns:
Current mode string.

Examples:

`CAPTure:MODE ONES` *one-shot capture*

`CAPTure:TRIGger(?) <b>`

Enables or disables trigger-start mode. When enabled, capture begins on the next rising edge of the channel-A reference input after arming.

Arguments:
`<b>` — `ON` or `OFF`

Returns:
Current state.

Examples:

`CAPTure:TRIGger ON` *arm on trigger*

`CAPTure:STARt`

Arms and starts the capture. If trigger mode is enabled, the capture enters the `ARMED` state and waits for a trigger. Otherwise, capture begins immediately.

Examples:

`CAPTure:STARt` *begin capture*

CAPTure:STOP

Stops an active capture. Data captured so far is retained in the buffer.

Examples:

`CAPTure:STOP` *halt capture*

CAPTure:STATus?

Queries the current capture state.

Returns:
`IDLE`, `ARMED`, `RUNNING`, or `FULL`.

Examples:

`CAPTure:STATus?` *check capture state*

CAPTure:POINts?

Queries the number of data points currently stored in the capture buffer.

Returns:
Integer.

Examples:

`CAPTure:POINts?` *how many points captured so far*

CAPTure:DATA? <first>,<count>

Reads captured data from the buffer. Returns an IEEE 488.2 definite length binary block containing *count* samples starting at index *first*. Each sample is one IEEE 754 single-precision float (4 bytes) per configured channel.

Arguments:
`<first>` — starting sample index (0-based)
`<count>` — number of samples to read

Returns:
Binary block with `count` × `channels` × 4 bytes.

Examples:

`CAPTure:DATA? 0,1000` *read first 1000 samples*

5.9 STReam Subsystem

STReam:ENAble(?) <b>

Starts or stops Ethernet data streaming. When enabled, the SR835 sends a continuous stream of measurement data to the host that issued the command.

Arguments:
<b> — ON or OFF

Returns:
Current state.

Examples:

STReam:ENAble ON *start streaming*
STReam:ENAble OFF *stop streaming*

STReam:CONFig(?) <c>

Sets or queries the streamed channel configuration.

Arguments:
<c> — X, XY, RT, or XYRT

Returns:
Current configuration.

Examples:

STReam:CONFig XYRT *stream all four quantities*

STReam:RATE(?) <n>

Sets or queries the stream rate divisor exponent. The actual stream rate is the base demodulation rate divided by 2^n.

Arguments:
<n> — integer, 0–15

Returns:
Current divisor exponent.

Examples:

STReam:RATE 4 *stream at base_rate / 16*

STReam:FORMat(?) <f>

Sets or queries the binary data format for streaming.

Arguments:
<f> — FLOAT32 (IEEE 754 single-precision) or INT16 (signed 16-bit, scaled to sensitivity)

Returns:
Current format string.

Examples:

STReam:FORMat FLOAT32 *use 32-bit floats*

STReam:PACKet(?) <b>

Sets or queries the packet size in bytes.

Arguments:
<b> — 128, 256, 512, or 1024

Returns:
Current packet size.

Examples:

STReam:PACKet 1024 *1024-byte packets*

STReam:PORT(?) <p>

Sets or queries the UDP/TCP destination port for streaming.

Arguments:
<p> — integer, 1024–65535

Returns:
Current port number.

Examples:

STReam:PORT 5555 *stream to port 5555*

STReam:ENDian(?) <e>

Sets or queries the byte order for streamed binary data.

Arguments:
<e> — LITTle or BIG

Returns:
Current byte order.

Examples:

`STReam:ENDian LIT` *little-endian byte order*

5.10 SCAN Subsystem

The SR835 can sweep (scan) an internal parameter while the lock-in measures the response. Available scan targets include channel frequencies, amplitudes, DC offsets, and auxiliary outputs.

`SCAN:TARGet(?) <t>`

Sets or queries the parameter to sweep during a scan.

Arguments:
`<t>` — `FREQ:A`, `FREQ:B`, `AMPL:A`, `AMPL:B`, `DC:A`, `DC:B`, `AUX1`, or `AUX2`

Returns:
Current target string.

Examples:

`SCAN:TARGet FREQ:A` *sweep channel-A frequency*

`SCAN:STARt(?) <v>`

Sets or queries the sweep start value. The unit and range depend on the scan target.

Arguments:
`<v>` — numeric value with optional unit suffix

Returns:
Start value in base units.

Examples:

`SCAN:STARt 100` *start at 100 Hz (if target is FREQ)*

`SCAN:STOP(?) <v>`

Sets or queries the sweep stop value.

Arguments:
`<v>` — numeric value with optional unit suffix

Returns:
Stop value in base units.

Examples:

`SCAN:STOP 100 KHZ` *stop at 100 kHz*

`SCAN:SHAPe(?) <s>`

Sets or queries the sweep shape.

Arguments:
`<s>` — `LINear` or `LOGarithmic`

Returns:
Current shape string.

Examples:

`SCAN:SHAPe LOG` *logarithmic sweep*

`SCAN:DURation(?) <t>`

Sets or queries the total sweep duration.

Arguments:
`<t>` — 0.001 to 99999 seconds

Returns:
Duration in seconds.

Examples:

`SCAN:DURation 60` *60-second sweep*

`SCAN:END(?) <e>`

Sets or queries the behavior at the end of a sweep.

Arguments:
`<e>` — `ONCE` (stop), `REPeat` (restart from beginning), or `UPDOwn` (reverse direction)

Returns:
Current end behavior.

Examples:

`SCAN:END REP` *repeat continuously*

SCAN:RUN

Starts the scan. The swept parameter begins moving from the start value toward the stop value.

Examples:

`SCAN:RUN` *start the sweep*

SCAN:STOP:IMMediate

Halts the scan immediately. The swept parameter remains at its current value.

Examples:

`SCAN:STOP:IMMediate` *abort scan*

SCAN:STATus?

Queries the scan state.

Returns:
`IDLE`, `RUNNING`, or `PAUSed`.

Examples:

`SCAN:STATus?` *check scan progress*

5.11 FFT Subsystem

The SR835 includes a real-time FFT spectrum analyzer for diagnosing signal and noise. The FFT can tap the signal at three points in the processing chain: raw ADC input, after the mixer, or after the demodulator low-pass filter.

FFT:SOURce(?) <s>

Sets or queries the signal tap point for the FFT analyzer.

Arguments:
`<s>` — `ADC` (raw input after preamp), `MIXer` (after PSD, before filter), or `FILTer` (demodulated output after low-pass filter)

Returns:
Current source string.

Examples:

`FFT:SOURce ADC` *analyze raw ADC signal*

`FFT:CHANnel(?) <n>`

Selects or queries which channel is analyzed by the FFT.

Arguments:
`<n>` — `A` or `B`

Returns:
Current channel.

Examples:

`FFT:CHANnel A` *analyze channel A*

`FFT:AVERage(?) <n>`

Sets or queries the number of RMS averages for the FFT display.

Arguments:
`<n>` — 1, 3, 10, 30, or 100

Returns:
Current average count.

Examples:

`FFT:AVERage 10` *10 RMS averages*

`FFT:SCALe(?) <s>`

Sets or queries the vertical scale of the FFT display.

Arguments:
`<s>` — dB per division: 1, 2, 5, 10, 20, 50, 100, or 200

Returns:
Current scale in dB/div.

Examples:

`FFT:SCALe 10` *10 dB/div*

`FFT:REFLevel(?) <r>`

Sets or queries the reference level (top of screen) for the FFT display.

Arguments:
`<r>` — −200 to +40 dBV

Returns:
Reference level in dBV.

Examples:

`FFT:REFLevel -20` *set -20 dBV reference*

`FFT:DATA?`

Returns the current FFT spectrum as 1024 comma-separated values in dBV, from DC to the Nyquist frequency.

Returns:
1024 comma-separated dBV values.

Examples:

`FFT:DATA?` *read full spectrum*

5.12 SYSTem Subsystem

`SYSTem:LINE(?) <f>`

Sets or queries the mains line frequency, used for notch filter tuning.

Arguments:
`<f>` — `50` or `60` (Hz)

Returns:
Current setting.

Examples:

`SYSTem:LINE 60` *set 60 Hz line frequency*

`SYSTem:TIMEbase(?) <s>`

Sets or queries the 10 MHz timebase source.

Arguments:
`<s>` — `INTernal` or `EXTernal`

Returns:
Current source.

Examples:

`SYSTem:TIMEbase INT` *use internal timebase*

`SYSTem:ETHernet:MODE(?) <m>`

Sets or queries the network IP configuration mode.

Arguments:
`<m>` — `DHCP` or `STATic`

Returns:
Current mode.

Examples:

`SYSTem:ETHernet:MODE DHCP` *use DHCP*

`SYSTem:ETHernet:IP(?) <addr>`

Sets or queries the static IP address. Changes take effect on the next power cycle.

Arguments:
`<addr>` — IPv4 dotted-quad string (e.g. `192.168.1.100`)

Returns:
Current IP address string.

Examples:

`SYSTem:ETHernet:IP 192.168.1.100` *set static IP*

`SYSTem:ETHernet:MAC?`

Queries the Ethernet MAC address (read-only, factory-set).

Returns:
MAC address string (e.g. `00:1A:2B:3C:4D:5E`).

Examples:

`SYSTem:ETHernet:MAC?` *read MAC address*

`SYSTem:KCLick(?) <b>`

Enables or disables the key-click sound on the front panel.

Arguments:
<b> — ON or OFF

Returns:
Current state.

Examples:

`SYSTem:KCLick OFF` *disable key click*

SYSTem:ALARm(?) <b>

Enables or disables the overload alarm sound.

Arguments:
<b> — ON or OFF

Returns:
Current state.

Examples:

`SYSTem:ALARm ON` *enable overload alarm*

SYSTem:ERRor:CLEar

Clears the ERR LED and the error queue.

Examples:

`SYSTem:ERRor:CLEar` *clear error indicators*

SYSTem:VERSion?

Returns the firmware version string.

Returns:
Version string (e.g.
`1.0.0`).

Examples:

`SYSTem:VERSion?` *read firmware version*

SYSTem:TRIP:CLEar

Clears a latched hardware trip condition. The hardware fault must be resolved before the trip can be cleared.

Examples:

`SYSTem:TRIP:CLEar` *clear latched trip*

`SYSTem:LOCal`

Returns the SR835 to local mode, re-enabling the front panel. This is the counterpart to `SYSTem:REMote`.

Examples:

`SYSTem:LOCal` *return to local mode*

`SYSTem:REMote`

Places the SR835 in remote mode, locking out the front panel. The user can still press the Local key to return to local mode.

Examples:

`SYSTem:REMote` *enter remote mode*

`SYSTem:RWLock`

Places the SR835 in remote mode with local lockout. The front panel is completely disabled, including the Local key. Only `SYSTem:LOCal` can return control.

Examples:

`SYSTem:RWLock` *remote with local lockout*

5.13 STATus Subsystem

`STATus:OPERation?`

Queries the Operation Status event register. Reading clears the register.

Returns:
Integer bitmask.

Examples:

`STATus:OPERation?` *read and clear operation status*

`STATus:OPERation:ENAble(?) <mask>`

Sets or queries the Operation Status enable mask. When (event & enable) is non-zero, bit 7 in the Status Byte is set.

Arguments:
`<mask>` — integer, 0–65535

Returns:
Current enable mask.

Examples:

`STATus:OPERation:ENAble 255` *enable lower 8 bits*

`STATus:QUEStionable?`

Queries the Questionable Status event register. Reading clears the register.

Returns:
Integer bitmask.

Examples:

`STATus:QUEStionable?` *read and clear questionable status*

`STATus:QUEStionable:ENAble(?) <mask>`

Sets or queries the Questionable Status enable mask. When (event & enable) is non-zero, bit 3 in the Status Byte is set.

Arguments:
`<mask>` — integer, 0–65535

Returns:
Current enable mask.

Examples:

`STATus:QUEStionable:ENAble 255` *enable lower 8 bits*

`STATus:LIA?`

Queries the lock-in amplifier status word. Each bit indicates a specific instrument condition (overload, unlock, etc.). Reading clears the register.

Returns:
Integer bitmask (see Lock-In Status Word table in the Programming chapter).

Examples:

`STATus:LIA?` *read and clear LIA status*

`STATus:LIA:ENAble(?) <mask>`

Sets or queries the LIA status enable mask. When (LIA & enable) is non-zero, bit 0 in the Status Byte is set.

Arguments:
`<mask>` — integer, 0–255

Returns:
Current enable mask.

Examples:

`STATus:LIA:ENAble 255` *enable all LIA status bits*

`STATus:ERRor?`

Queries the error status byte. Reading clears the register.

Returns:
Integer bitmask (see Error Status Byte table in the Programming chapter).

Examples:

`STATus:ERRor?` *read and clear error status*

`STATus:ERRor:ENAble(?) <mask>`

Sets or queries the error status enable mask. When (error & enable) is non-zero, bit 2 in the Status Byte is set.

Arguments:
`<mask>` — integer, 0–255

Returns:
Current enable mask.

Examples:

`STATus:ERRor:ENAble 255` *enable all error bits*

`STATus:PRESet`

Resets all status enable registers to their default values (zero). Event registers are not affected.

Examples:

`STATus:PRESet` *clear all enable masks*

5.14 DISPlay Subsystem

`DISPlay:LAYout(?) <l>`

Sets or queries the screen layout.

Arguments:
`<l>` — `FULL` (single pane), `DUAL` (two panes), `QUAD` (four panes), `CHART` (strip chart), `FFT` (spectrum), or `SCAN` (scan display)

Returns:
Current layout string.

Examples:

`DISPlay:LAYout DUAL` *two-pane display*

`DISPlay:CHANnel<n>:QUANtity(?) <q>`

Sets or queries the quantity shown in display pane *n*.

Arguments:
`<n>` — pane number, 1–4 (depending on layout)
`<q>` — `X`, `Y`, `R`, `THETa`, `XNOise`, `YNOise`, or `FREQuency`

Returns:
Current quantity string.

Examples:

`DISPlay:CHANnel1:QUANtity R` *show R in pane 1*

`DISPlay:BLANk(?) <b>`

Blanks or unblanks the display. Blanking the display can reduce electrical noise radiated by the screen.

Arguments:
`<b>` — `ON` (blank) or `OFF` (normal display)

Returns:
Current blanking state.

Examples:

`DISPlay:BLANk ON` *blank the screen to reduce noise*

`DISPlay:CAPTure`

Captures the current screen image and saves it to USB storage. A USB drive must be connected to the front-panel USB-A port.

Examples:

`DISPlay:CAPTure` *save screenshot to USB*